Industrial Policy and Jobs: The Role of Production Networks and Labor–Materials Substitution

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Abstract

I study how pro-competition industrial policies affect labor demand. Importantly, I distinguish between direct effects on targeted sectors, where prices and markups decline, and indirect effects that propagate through both input-output linkages and heterogeneous labor-material substitution elasticities. Input-output linkages determine where price reductions propagate, while the elasticity of substitution between labor and materials governs firms' labor response to lower material prices. Labor demand is amplified when labor and materials are complements and reduced when substitutes. I quantify these channels in the context of a major pro-competition policy in India, which lifted entry restrictions in about 500 product-markets. I estimate a network model with oligopolistic competition, variable markups, and sectoral-level labor-materials elasticity of substitution. The labor–materials substitution channel explains over one third of the aggregate labor gains from deregulation, more than the standard markup-compression channel. For every 100 jobs created in deregulated sectors, 57 survive, and 43 are destroyed through reallocation. Without the labor–materials substitution channel, 60 would have been destroyed. Crucially, heterogeneity in labor-materials elasticity of substitution matters for sectoral policy evaluation: for every 100 jobs created in deregulated sectors, a common-elasticity model assigns about 23 to the wrong sectors. The key policy takeaway is that employment effects depend on which sectors buy the goods the reform makes cheaper. Counterfactual exercises show that reforms generating similar output gains differ substantially in job gains or losses depending on how they are targeted.

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Industrial policy has regained prominence in recent decades as governments seek to promote growth, competition, and employment indicators (Juhász et al., 2024). The evaluation of these policies are often in terms of output and job gains, and are made on the sectors directly targeted (Criscuolo et al., 2019; Barwick et al., 2025; Cingano et al., 2025). However, these policies affect more than targeted sectors. In network economies, industrial policies that change prices and costs propagate to downstream sectors. Hence, evaluating aggregate effects requires fully accounting for indirect effects that propagate through production networks[1].

In this paper, I study how technology heterogeneity amplifies or dampens the labor responses to Industrial Policy shocks, on targeted and untargeted industries. I study this in a context where Input-Output linkages are present and the elasticity of substitution between labor and intermediate inputs varies across sectors. Particularly, the elasticity of substitution between labor and intermediate inputs determines how firms optimally adjust their input mix towards more or less labor, upon a change in intermediate input prices (henceforth, the substitution channel). When intermediate inputs become cheaper, complement firms hire more and substitute firms replace workers with materials. The heterogeneity in this elasticity is commonly absent in network economies. I show that incorporating that heterogeneity in a network economy has implications for the evaluation of the propagation effects of industrial policies on employment outcomes, even though it does not affect the effects on output.

The substitution channel is studied relative to two additional mechanisms through which industrial policy operates. First, a reallocation channel captures how demand shifts across sectors as the policy changes relative sectoral prices: final consumers substitute toward the deregulated sectors, downstream firms shift their input purchases toward the now-cheaper suppliers, and the input–output network propagates and amplifies these shifts. Second, a markup channel captures the pro-competitive effect of the policy: entry lowers markups and reallocates market share away from high-markup firms. A larger share of each unit of revenue is paid to variable inputs, raising labor demand.

I study the relative importance of these mechanisms, with an special focus on the substitution channel, in the context of the Indian Small-Scale dereservation reform, a pro-competitive industrial

[1] Carvalho and Tahbaz-Salehi (2019) offer a comprehensive review of production networks literature.

policy [2]. This reform removed entry barriers for large firms in about 500 product lines, by eliminating a capital cap that prevented them to operate in such markets. In my paper, the policy serves for two purposes. First, it lets me generate a set of empirical findings that show that industrial policy effects on labor are driven by the technology distribution. Second, it informs the structural model about the concentration, expansion and entry effects caused by the policy.

Using the policy as an exogenous shock, I find that firms in non-targeted sectors that purchase inputs that were targeted by the reform increase their labor demand. These employment gains are driven by sectors where labor and intermediate inputs are complements, while substitute sectors show no significant response. This is consistent with the heterogeneous input mix responses that are shaped by the heterogeneous labor and intermediate input substitution pattern. Additionally, I report that firms that are affected through their output market exhibit significant declines in product-level markups in targeted sectors and positive labor demand effects. This result is consistent with the competitive pressure introduced by the reform. The predictions on labor demand are theoretically consistent, as removing markup distortion generates incentives for firms to grow larger, expand production, and hire more workers.

After establishing these empirical results, I build a network economy model with variable markups, firms competing in oligopolistic markets, and heterogeneous elasticities of substitution between labor and intermediate inputs. The policy is modelled by removing capital constraints for capped incumbents and potential entrants. The entry is exogenous as in Edmond et al. (2015). The model allows me to account for input-output linkages and captures general equilibrium effects on labor demand that cannot be recovered from estimates that restrict the analysis to targeted sectors only, as a Difference in Differences model. It also lets me cleanly decompose the policy-induced labor-demand gains into the three previously discussed channels: a reallocation channel, a markup channel and the substitution channel. This decomposition lets me quantify the contribution of heterogeneous production technologies to aggregate employment.

To incorporate the heterogeneous elasticity of substitution between labor and intermediate inputs, I exploit the firm optimality conditions to generate an estimable equation that requires observing intermediate input prices and wages, as in Peter and Ruane (2025) and Atalay (2017). I

[2]This policy has been extensively studied. Evaluation has shown that it generated output and labor expansion, firm entry and markup declines (Martin, Nataraj and Harrison, 2017; García-Santana and Pijoan-Mas, 2014; Galle, 2018)

show that more than 65% of sectors use labor and intermediate inputs as complements, consistent with recent India-based findings that economy-wide substitution aligns with firms using labor and materials as complements. I also show substantial heterogeneity in this elasticity across narrowly defined sectors, with point estimates ranging from 0.36 to 1.73. This suggests a pattern moving from sectors that use labor and intermediate inputs to sectors that can easily substitute one for the other.

In the first model result, I find that for every 100 jobs created in targeted sectors, the reform generates a net increase of 57 jobs economy-wide. The net difference captures the general equilibrium adjustments. Particularly, I show that demand reallocation across sectors generates employment losses in non-targeted sectors of 60 jobs for every 100 jobs created in targeted sectors. However, labor-intermediate input complementarities partially offset the losses, recovering 17 jobs. Notably, economy-wide, the substitution channel accounts for over a third of total net job creation. This exceeds the contribution of markup reduction, which accounts for about 18% of the total net effect. This result has a key policy implication: the economy's technology structure matters for policy design and evaluation. Ignoring labor-intermediate inputs substitution patterns can lead to substantial mismeasurements of the policy's effect.

Afterward, I identify the role of labor-intermediate input substitution levels assumptions. I recalibrate the model under scenarios where I assume a Cobb-Douglas production function, or I allow common elasticities of substitution commonly assumed in the literature. I find that the Cobb-Douglas case understates aggregate job creation by 30% relative to the benchmark. Conversely, assuming a homogeneous elasticity of substitution of 0.7 overstates aggregate job creation by about 25%. Finally, using the aggregate-level point estimate reduces the overstatement, though it remains a non-negligible 14%.

The latter counterfactuals highlight the importance of the elasticity *level* for the aggregate results. However, they do not reveal the importance of *sectoral heterogeneity*. To highlight the importance of heterogeneity, I go beyond the aggregate results and show that setting a common economy-wide elasticity of substitution roughly quintuples the net job gains in substitute sectors relative to the baseline heterogeneous-elasticity model. Those sectors account for about one third of the economy's total employment. Moreover, the common-elasticity version understates job gains in complement

sectors by about 5%. Altogether, the common-elasticity model misattributes which sectors gain jobs, even though aggregate effects seem close to the heterogeneous model version. This result is crucial for policy design, as it underscores the importance of considering not only network linkages but also technology heterogeneity across the network to design policies that achieve the intended consequences in targeted and untargeted sectors.

To illustrate the last point, my last exercise proposes two alternative policies that, by design, have similar output effects and affect a similar share of employment in the economy. The only difference between the policies is the elasticity of substitution of the reached sectors through the network. The key result is that, under a common elasticity of substitution, both reforms would create the same number of jobs. Under the estimated heterogeneous elasticity, the reform that reaches complements through the networks generates 25% fewer jobs. In comparison, the reform that reaches substitutes creates 45% fewer jobs, relative to the common-elasticity model.

Related Literature: My main contribution is to the growing literature on technology's effects on labor outcomes. One strand has studied technology's complementary role in aggregate labor outcomes, labor share, and welfare. Bonilla-Mejía et al. (2026) study the employment consequences of trade liberalization under heterogeneous labor–materials substitution elasticities, while Castro-Vincenzi and Kleinman (2026) show how changes in intermediate-input prices affect the labor share of value added. My paper builds on this idea by studying another source of input-price changes: industrial policy that facilitates entry and expansion. This source generates richer interactions through endogenous price and markup adjustments driven by product-market competition. This lets me quantify the relative contributions of markup compression, demand reallocation, and labor–materials substitution to industrial policy's employment effects.

Second, I contribute to the literature that exploits the CES structure between labor and intermediate inputs. One strand uses this structure to recover labor-augmenting productivity and to study its implications for labor demand and the labor share (Chen, 2017; Zhang, 2019). Another allows the elasticity of substitution to vary across sectors to discipline the model's responses: Bonilla-Mejía et al. (2026) recover three elasticities, for manufacturing, services, and agriculture, targeted to reproduce observed labor-demand responses. I complement this work by estimating the elasticities directly across narrowly defined sectors, using firm-level production data and instrumental vari-

ables, building on Atalay (2017) and Peter and Ruane (2025). These externally estimated elasticities, rather than targeted ones, then discipline the model's implied labor responses.

This approach captures substantial technological heterogeneity within manufacturing and separates production-technology estimation from observed labor-demand responses to policy shocks. In this way, I follow the Industrial Organization tradition of estimating sector-specific production-function parameters and recognizing structural technology differences across sectors (Blum et al., 2024; Gandhi et al., 2020; Oberfield and Raval, 2021; De Loecker and Warzynski, 2012). I also follow recent advances that reconcile structural granular estimation from the Industrial Organization literature with macro models (Holguin et al., 2026; Morlacco, 2020; De Loecker et al., 2020).

Finally, I contribute to the literature on shock propagation through production networks. Recent literature has focused on identifying productivity shock propagation (Acemoglu et al., 2012; Barrot and Sauvagnat, 2016; Grassi, 2017; Baqaee and Farhi, 2019; Carvalho et al., 2021), industrial policy propagation (Liu, 2019; Lane, 2025), and other distortions such as quotas and wedges (Jones, 2013; Bigio and La'O, 2020; Baqaee and Farhi, 2020; Baqaee and Sangani, 2025). In my framework, I show that network position and shock effects on output are insufficient to evaluate employment consequences when the elasticity of substitution between labor and intermediate inputs is heterogeneous, with important implications for industrial policy design and evaluation.

1 Empirical Evidence

1.1 De/Reservation Policy

To study the mechanisms proposed in this paper, I use the Indian Small-Scale Industries dereservation policy. This was India's dismantling of a small-scale reservation policy in product markets. The reservation policy, which started in 1967, aimed to protect small firms from competition from large firms. To do so, the Indian government proposed a list of products reserved for production by small firms exclusively, creating an artificial entry barrier for large firms in around 500 product-spaces. In practice, the entry restriction operated through a threshold in Plant and Machinery capital that firms could not surpass to participate in the restricted markets. From 1997 to 2015, India began

the dereservation process by dismantling these restrictions. The government unlisted previously reserved products in a staggered way, starting with a conservative number and peaking between 2001 and 2008 (Martin et al., 2017). A strand of research has evaluated this policy intensively, evidencing positive outcomes consistent with pro-competitive effects (Martin et al., 2017; Galle, 2018; Boehm et al., 2022)

1.2 Data

The primary data set used in this paper is the Annual Survey of Industries from 2001 to 2008. ASI covers all manufacturing plants in India with ten or more workers. The sampling design in this survey works as follows: from 2001 to 2004, all plants with 100 or more workers, or those located in the six (it was reduced to five by 2008) less industrially developed states, were eligible for a census and were required to fill the survey. A random sample is applied to the remaining plants.

This dataset has two main features that suit my research question. First, it records revenues/costs and quantities at the 5-digit level for output manufactured and inputs used in production. This aggregation level fits the level at which the dereservation policy operated. Due to the high granularity level that the data reports, and given that the treated unit was a product, I can identify when a firm was affected through its output market (i.e., through a product manufactured by the firm), or through the input market (i.e., through an input used by the firm for manufacturing). This distinction is key for our analysis, as it lets me identify the policy effects through two mechanisms: output market distortion reductions, and material price reductions through the propagation network. It also allows me to isolate the substitution channel and its differential effects on labor outcomes. Moreover, observing inputs used and output manufactured at the same aggregation level lets me separately identify when a firm was affected by the policy through input or output markets. Second, I can recover unit prices at the input and output levels. This lets me build firm-level intermediate input prices and estimate the sectoral elasticity of substitution between labor and materials. Moreover, output unit prices and quantities allow me to obtain markup estimates that bypass the identification challenge described by Bond et al. (2021) when revenue-based production functions are used to recover markups.

1.3 Empirical Evidence

In this section, I provide a set of empirical findings that motivate my modeling choices and illustrate the channels studied.

Finding 1: Heterogeneous Elasticity of Substitution between labor and materials

To estimate the elasticity of substitution between labor and materials, I use the relationship implied by the optimality conditions of a cost-minimizing firm that combines labor and materials using a CES production function. I estimate equation 1.1 using firm-level data. P_{it}^{M} denotes a Törnqvist firm-level input price index, while M_{it} is the quantity index such that

$$P_{it}^{M} M_{it} = \sum_{j} p_{ijt} m_{ijt}$$

where j denotes individual input used by the firm, and p^m and m denote the price and quantity reported by the firm i. w_{it} denotes the average wage paid by the firm in year t, and it is calculated as the ratio of the total wage bill [3] to the average number of employees. I include firm-level controls such a as firm age, sector, and location, denoted by the vector $\mathbf{X_{it}}$

Naturally, estimating equation 1.1 using OLS would induce bias due to the correlation between unobserved demand shocks ε_{it} and relative prices. I follow Atalay (2017) and Peter and Ruane (2025), and adopt an instrumental variable approach to estimate η_s. I follow Orr (2022) and define an instrument that affects relative labor-intermediate input expenditures through the cost side. For that, I capture the intermediate input price and wage variation that occurs at firms located in close districts and operate in the same sector as firm i.

$$\log\left(\frac{P_{it}^{M} M_{it}}{w_{it} L_{it}}\right) = \alpha + (\eta_s - 1)\log\left(\frac{w_{it}}{P_{it}^{M}}\right) + \mathbf{X}_{it}'\gamma + \delta_t + \varepsilon_{it}. \tag{1.1}$$

Let $d(i)$ denote the district in which firm i operates and $\mathcal{D}^{200}(d)$ the set of districts whose centroids lie within 200 km of the centroid of district d, excluding d itself. Let $\mathcal{I}_{st}^{200}(d)$ denote the set of firms operating in sector s and year t that are located in a district in $\mathcal{D}^{200}(d)$. The wage and

[3] Includes salary plus benefits and welfare expenditures

material-price instruments are the average log wage and the average log material price of these firms,

$$Z_{ist}^{w,200} = \frac{1}{N_{st}^{200}(d(i))} \sum_{k \in \mathcal{I}_{st}^{200}(d(i))} \log w_{kt}, \qquad Z_{ist}^{M,200} = \frac{1}{N_{st}^{200}(d(i))} \sum_{k \in \mathcal{I}_{st}^{200}(d(i))} \log P_{kt}^{M},$$

and the relative-price instrument is $Z_{ist}^{200} \equiv Z_{ist}^{w,200} - Z_{ist}^{M,200}$. As an additional wage instrument, I use the local labor-market wage, the average log wage of firms in the same district and year that operate in a different 2-digit industry (Oberfield and Raval, 2021).

The distance instruments vary only by district, sector, and year, so they lose power within individual sectors (Appendix E). For the sector elasticities, I follow Peter and Ruane (2025) and estimate equation 1.1 in long differences between a base year t_0 and $t_1 = t_0 + h$, for horizons $h = 1, \ldots, 4$,

$$\Delta \log \left(\frac{P_i^M M_i}{w_i L_i} \right) = (\eta_s - 1) \, \Delta \log \left(\frac{w_i}{P_i^M} \right) + \delta_{h,t_0} + \lambda_{h,n(i)} + \Delta \varepsilon_i,$$

where δ_{h,t_0} are horizon-by-base-year effects and $\lambda_{h,n(i)}$ are horizon-by-sector effects for the firm's base-year 3-digit sector. I instrument the change in the relative price with a shift-share of national input-price changes. For each input j, let $\bar{p}_{jt}^{-d}$ be the mean log unit value paid for j in year t by the set $\mathcal{B}_{jt}^{-d}$ of firms located outside district d, and let s_{ij,t_0} be the share of input j in firm i's domestic materials expenditure in the base year. The instrument is

$$\bar{p}_{jt}^{-d} = \frac{1}{N_{jt}^{-d}} \sum_{k \in \mathcal{B}_{jt}^{-d}} \log p_{kjt}, \qquad Z_i = \frac{\sum_j s_{ij,t_0} \, \Delta \bar{p}_j^{\,-d(i)}}{\sum_j s_{ij,t_0}},$$

where $\Delta \bar{p}_j^{\,-d(i)} = \bar{p}_{jt_1}^{-d(i)} - \bar{p}_{jt_0}^{-d(i)}$. Z_i is the change in the firm's material price if it kept its base-year basket and paid the national price changes. The firm's actual material-price change $\Delta \log P_i^M$ uses the same price changes with Törnqvist weights, the average of its base-year and end-year shares.

The identifying assumption is that national price changes of the inputs in a firm's base-year basket are uncorrelated with the change in its unobserved factor bias $\Delta \varepsilon_i$, conditional on the fixed effects. Two features of the design support it. First, the price changes exclude the firm's own district, so they do not reflect the firm's own purchases or local demand and productivity shocks that also

move its input mix. Second, the shares are fixed at t_0, so the instrument does not respond to the firm's input choices after the base year. Second, differencing removes time-invariant differences in technology. Identification therefore comes from firms in the same sector that buy different baskets and so face different national price changes. The assumption would fail if national input-price changes reflected technology shocks specific to the firms that buy those inputs, beyond what the sector effects absorb.

Table 1 presents the pooled estimates of both designs. As expected, the biased OLS estimator biases the result towards one. The endogeneity correction results in a coefficient that describes an economy where labor and materials are complements.

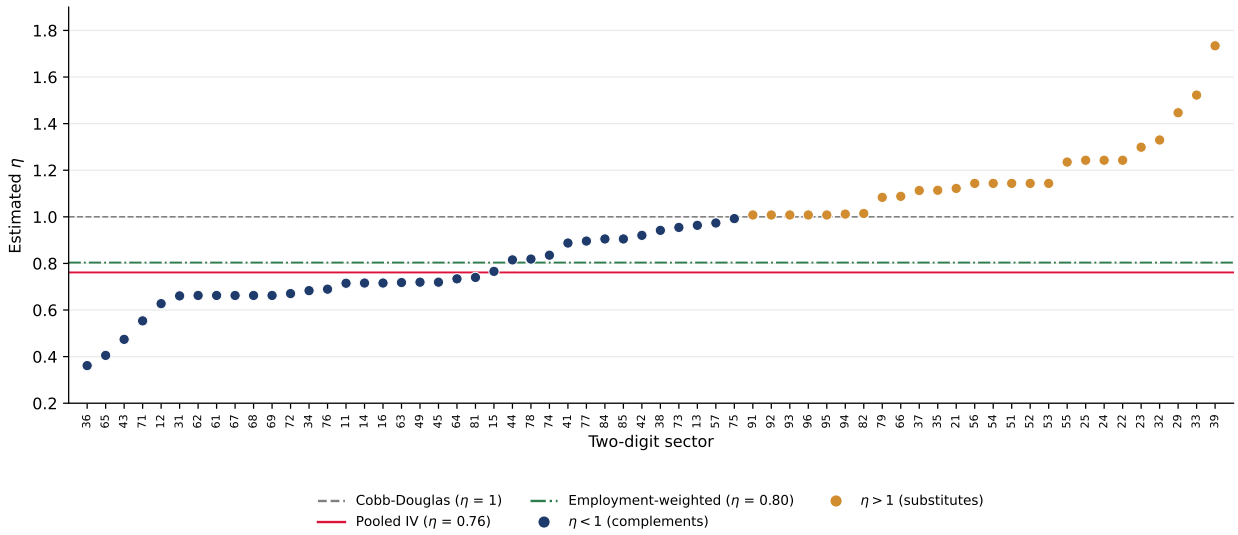
Table 1: Pooled Labor-Intermediate Inputs Elasticity

	Levels		Long differences
	OLS	IV	IV
	(1)	(2)	(3)
η	0.914	0.758	0.761
	(0.006)	(0.128)	(0.025)
Instrument	–	Z_{ist}^{200}	Z_i
First-stage F	–	31.0	4,147.1
Observations	92,609	92,609	64,085

Notes: Columns (1) and (2) estimate equation 1.1 in log levels on firm-years. Column (2) instruments $\log(w_{it}/P_{it}^M)$ with Z_{ist}^{200}, the average log wage minus the average log material price of firms in the same 3-digit sector and year located in other districts within 200 km. Both columns include 3-digit sector $\times$ year fixed effects and state fixed effects, and control for firm age and type of organization; standard errors are clustered by district. Column (3) estimates equation E.1 in long differences on firm-pairs, stacking horizons of 1 to 4 years, with horizon $\times$ base-year and horizon $\times$ 3-digit sector fixed effects, and instruments the change in the relative price with the shift-share Z_i of equation E.2, the instrument used for the sector elasticities; standard errors are clustered by firm. Standard errors in parentheses. First-stage F is the Kleibergen–Paap Wald statistic.

A second important fact emerges when I confine the attention to the sector-level elasticity estimation of equation 1.1. Figure 1 summarizes the results. I find a pronounced heterogeneity in the elasticity of substitution when estimated at the 2-digit sector level. In particular, I find 36 sectors classified as complements and 26 as substitutes. Elasticities range from 0.36 to 1.73. This result is key, as it suggests that changes in intermediate input prices should have heterogeneous effects on labor outcomes across sectors. This is particularly relevant in network economies, as price shocks propagate differently depending on the economy's production-technology distribution, implying that measures of aggregate labor outcomes depend on assumptions about that distribution.

Figure 1: Elasticity of Substitution between Labor and Materials, 2-Digit Sectors



Notes: Point estimates of η_s from equation E.1, horizons 1–4 stacked, instrumented with the shift-share Z_i (equation E.2), ranked by value. For 25 two-digit sectors, accounting for 2.3% of employment, the 2-digit estimate is unavailable, these sectors are assigned the estimate of their 1-digit sector. The dashed line marks the Cobb-Douglas case ($\eta = 1$). The pooled IV estimate is from the same specification without sector heterogeneity; the employment-weighted mean uses each 2-digit sector's employment as weights.

Finding 2: Heterogeneous Elasticity is associated with shocks effects on labor outcomes

The main theoretical implication of the first finding is that a shock to intermediate input prices affects firms' labor demand differently depending on whether sectors treat labor and materials as complements or substitutes. I test this theoretical relation by exploiting the granularity of the dereservation policy and my available data. In particular, the ASI dataset records the products a given firm produces and the inputs it uses at the same level as the policy-affected products. This distinction lets us identify whether a firm in a given year was affected because one of its production inputs was dereserved.

To empirically evaluate the interaction between labor demand changes and the substitutability between labor and materials, I estimate an event study as follows:

$$\log L_{it} = \sum_{k \in \mathcal{K}} \beta_k^C \left[\mathbb{1}\{\tau_{it} = k\} \times D_s^C \right] + \sum_{k \in \mathcal{K}} \beta_k^S \left[\mathbb{1}\{\tau_{it} = k\} \times D_s^S \right] + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1.2}$$

Where

$$D_s^S = \mathbb{1}\{\eta_s > 1\}, \qquad D_s^C = \mathbb{1}\{\eta_s < 1\}, \tag{1.3}$$

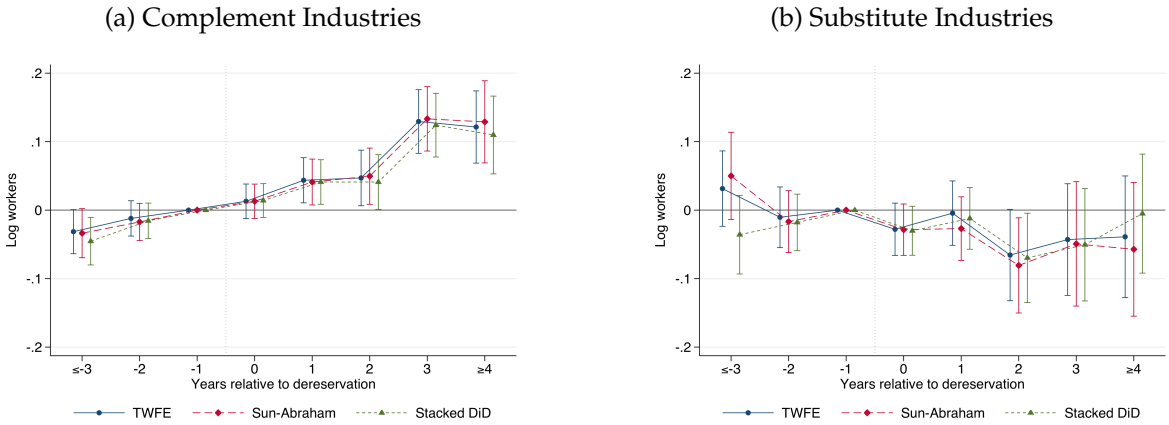
α_i and δ_t are firm and year fixed effects (the firm effect absorbs the firm's region and sector), and τ_{it} denotes event time relative to the year in which the firm's first input is dereserved. Appendix G shows the results under alternative treatment dates, including the main input dereservation date, or any input used at any period dereserved. The omitted period is $k = -1$, so each coefficient measures the change in employment relative to the year immediately preceding input dereservation. Because dereservation is staggered across years, I estimate equation 1.2 with three estimators: two-way fixed effects (TWFE); the interaction-weighted estimator of Sun and Abraham (2021), which interacts event time with the dereservation cohort; and a stacked difference-in-differences (Cengiz et al., 2019), which builds one stack per cohort over the window $[-3, 4]$ with never-treated firms and firms treated after the window as controls, and replaces α_i and δ_t with firm $\times$ stack and year $\times$ stack fixed effects. The coefficients β_k^C trace the employment response in sectors where $\eta_s < 1$, while β_k^S trace the corresponding response in sectors where $\eta_s > 1$. Thus, differences between β_k^C and β_k^S reveal whether the propagation of the input-side policy shock depends on the substitutability classification between labor and materials.

The results are presented in Figure 2. The event study design reveals that the theoretical prediction holds in the data. Particularly, upon increasing competition in markets from which firms source inputs, labor demand increases in firms that participate in sectors that are classified as complements. Conversely, I find no evidence of a significant effect in substitute markets. Importantly, by design, firms affected through the output side enter only in the years before their output dereservation. This allows me to avoid the potential conflation of the same policy affecting firms through distinct channels.

Finding 3: Firms affected through the output side also increase their labor demand

In variable markup economies, the distortions created by markup heterogeneity distort firm scale. High-markup firms produce less than their socially optimal output, and use fewer inputs as a consequence. Importantly, this result is common across oligopolistic competition models, and

Figure 2: Effects of Dereservation Policy



Notes: Event-study coefficients for log number of workers, with 95% confidence intervals; standard errors clustered by firm. Sectors are classified by the 2-digit η_s of Figure 1: complements ($\eta_s < 1$, 36 sectors) and substitutes ($\eta_s > 1$, 26 sectors). Firms whose own products were dereserved enter only in the years before their output dereservation. For each estimator, the complement and substitute coefficients come from a single regression. TWFE and Sun–Abraham include firm and year fixed effects, with leads binned at $k \leq -3$ and lags at $k \geq 4$; Sun–Abraham interacts event time with the dereservation cohort and aggregates with cohort weights. The stacked DiD builds one stack per dereservation cohort over the window $[-3, 4]$, with never-treated firms and firms treated after the window as controls, and includes firm $\times$ stack and year $\times$ stack fixed effects. Observations (TWFE and Sun–Abraham, singletons dropped): complements 46,873 firm-years (15,612 firms, 3,580 treated); substitutes 12,850 firm-years (4,186 firms, 1,071 treated). Stacked DiD: 222,514 and 58,714 firm-year-stack observations, respectively.

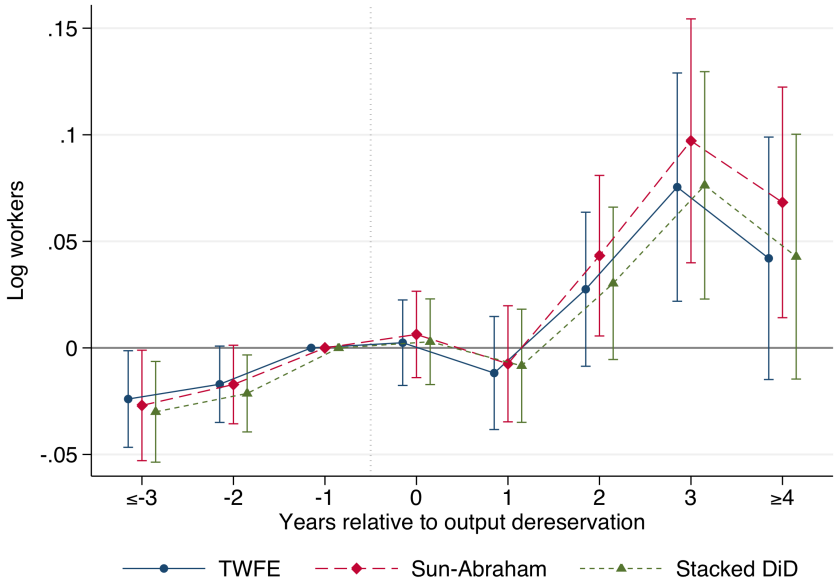
makes aggregate labor depend on aggregate markup levels (Edmond et al., 2015; Peters, 2020). Following the theoretical prediction, I estimate how firms affected through their manufactured products change their labor demand after the policy. Similar to the input dereservation, I estimate an event study described by equation 1.4.

$$\log L_{it} = \sum_{k \in \mathcal{K}} \beta_k^O \left[\mathbb{1}\{\tau_{it}^O = k\} \right] + \alpha_i + \delta_t + \varepsilon_{it}, \tag{1.4}$$

where τ_{it}^O denotes event time relative to the year in which the firm's first reserved product manufactured is dereserved, and α_i and δ_t are firm and year fixed effects. The sample, event-time set $\mathcal{K}$, and estimators are the same as in equation 1.2: TWFE, Sun and Abraham (2021), and the stacked difference-in-differences. Figure 3 summarizes the estimation results. My results show a positive and significant effect on labor demand when firms are affected through the output side. Firms affected through an input enter only in the years before their input dereservation, mirroring the input-side design. As before, this design avoids conflating the effects of firms affected by the same policy through different mechanisms. The positive result suggests that the policy generated

positive labor effects.

Figure 3: Event Study for Output



Notes: Event-study coefficients for log number of workers, with 95% confidence intervals; standard errors clustered by firm. Firms that make a reserved product other than their main product are excluded. Firms affected through an input enter only in the years before their input dereservation. TWFE and Sun–Abraham include firm and year fixed effects, with leads binned at $k \leq -3$ and lags at $k \geq 4$; Sun–Abraham interacts event time with the dereservation cohort and aggregates with cohort weights. The stacked DiD builds one stack per dereservation cohort over the window $[-3, 4]$, with never-treated firms and firms treated after the window as controls, and includes firm $\times$ stack and year $\times$ stack fixed effects. Observations (TWFE and Sun–Abraham, singletons dropped): 57,493 firm-years (19,347 firms, 3,026 treated). Stacked DiD: 271,035 firm-year-stack observations (19,249 firms, 2,928 treated).

The usual suspect mediating the effect between the policy and labor responses is markups. The dereservation policy acts as a pro-competitive policy, as it allows the entry of productive firms that were left out of the market due to their capital scale. The general implication is higher competitiveness in the economy, which translates into reduced markups. To test whether this mechanism operates in India after the policy, I use disaggregated information on quantities and unit prices available in the ASI. With this, I follow De Loecker et al. (2016) and estimate product-level markups using the production approach adapted to a multi-product firm context. I follow this method for two reasons. First, it yields markup estimates at the same disaggregation level as the policy operated. Second, estimating markups using quantity-based production functions helps me avoid recent criticism that revenue-based production functions cannot identify markups Bond et al. (2021). I discuss this extensively in Appendix D.

With markup estimation in place, I estimate equation 1.5. As before, β^O captures the average post-policy effect after controlling for pre-trends and fixed effects. Note that, in this specification, I include the subindex p, which denotes the product produced by firm i. Table 2 presents the results. In column 1, after the products are dereserved, the average markup within each product falls by about 10-13%. I conduct a series robustness exercises in column 2 and 3, and show that the results are relatively stable after considering different sets of fixed effects. In general, the results are as expected: dereservation policy induced a reduction in markups at the product level, which is consistent with a pro-competitive effect. Also, this result complement previous findings by Martin et al. (2017), that show that the reform induced entry and markets expansion, driven particularly by large entrants.

$$\log \mu_{ipt} = \beta^O Post^O_{ipt} + \sum_{k \in \mathcal{K}^{pre}} \gamma^O_k \left[\mathbb{1}\{\tau^O_{ipt} = k\} \right] + \alpha_p + \delta_t + \lambda_{s \times t} + \varepsilon_{ipt}, \tag{1.5}$$

Table 2: Effects of Output Dereservation on Firm Markups

	(1)	(2)	(3)
Post output dereservation	-0.096^{**}	-0.103^{***}	-0.139^{***}
	(0.038)	(0.038)	(0.051)
Product FE	Yes	Yes	Yes
NIC2 × year FE	Yes	Yes	Yes
State × year FE	No	Yes	No
Firm FE	No	No	Yes
Pre-trends p-value	0.149	0.174	0.419
Observations	214,294	214,294	187,611

Notes: The dependent variable is the log markup of the firm-product. Each regression includes indicators for 2–4 and 5 or more years before output dereservation; the pre-trends p-value tests that they are jointly zero. Standard errors clustered by firm-product in parentheses. Column (3) drops singleton groups. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

As in the previous designs, in the estimation of equation 1.5, I exclude all firms already affected through the input mechanism. My results are comparable to those of Galle (2018), although I use a broader sample that includes multi-product firms and entrants. Together, the results in figure 3 and table 2 suggest that the firms affected through output channel exhibit both a markup decline, and also labor demand growth, which will be captured by the model presented in the next sections.

1.4 Rationalizing the facts

In summary, my main empirical results are as follows. Pro-competition policies affect labor outcomes positively through two main channels. First, they reduce average markups in sectors that are targeted by the policy. Theoretically, such a decline is associated with firms operating at higher scales and using more inputs. This is validated by my data. Second, pro-competition policies positively affect labor demand for firms operating in non-targeted sectors, as long as they use inputs manufactured by targeted sectors. Moreover, the firms driving this positive effect are those that operate in sectors where labor and materials are complements. Finally, and crucially for what follows, I find that using an IV design and exploiting the data availability, elasticity of substitution across sectors is heterogeneous. This potentially affects how competition policies affect labor outcomes when the shock propagates through the network.

In the next section, I develop a static oligopolistic competition model with endogenous markups and production networks that captures the previously discussed empirical findings. Moreover, I build on García-Santana and Pijoan-Mas (2014) and add a layer that closely captures the dereservation policy. However, the mechanisms the model explains are not specific to this policy and can be generalized to any competition policy that incentivizes expansion or entry. The model's main goal is to decompose the policy effects on labor demand into a markup component and the substitution channel. The first accounts for the effects of pro-competition policies through reduced market distortion from reduced and recomposed markups. The second mechanism captures the portion of the effect explained by the economy's technology structure in terms of substitutability or complementarity between labor and materials. A clear understanding of how both mechanisms operate together and in isolation provides useful information for policymakers.

2 Model

I develop an oligopolistic competition model with endogenous markups and production networks. Firms combine inputs sourced from different industries. Crucially, I assume that each firm participating in sector s sources the same input recipe from other industries. I assume firms compete à la

Bertrand. I also assume firms pay operating costs each period and a one-shot sunk cost to enter.[4] I assume an exogenous number of incumbents operate in a baseline period with the dereservation policy in place. To model the dereservation policy, I assume that, in the baseline, a capital cap limits operation to firms below that level. The counterfactual policy eliminates the cap and studies how entry and expansion affect market structure, output, and input prices through the network. Finally, I trace how removing the cap affects labor outcomes through three main channels: reallocation, markup reduction, and substitution.

3 Environment

The economy has K three-digit sectors indexed by k. A sector is partitioned into *cells*: a reserved branch $R(k)$ and a never-reserved branch $U(k)$. Not every sector contains both. Write

$$\mathcal{S}(k) \subseteq \{R(k), U(k)\}, \qquad \mathcal{S}(k) \neq \emptyset, \qquad \mathcal{S} = \bigcup_{k=1}^{K} \mathcal{S}(k), \qquad |\mathcal{S}| = S \tag{3.1}$$

so that $S < 2K$, and let $\mathcal{N}_s$ denote the firms active in cell $s \in \mathcal{S}$ and $k(s)$ the sector containing s.[5]

3.1 Final demand

We assume that, for producing the final good Y, sectors s are aggregated using a constant-elasticity-of-substitution technology with elasticity parameter θ_f. $\Gamma(s)$ denotes a final-demand weight, and $C(s)$ denotes the final consumption of cell s good.

$$Y = \left(\sum_{s \in \mathcal{S}} \Gamma(s)^{\frac{1}{\theta_f}} C(s)^{\frac{\theta_f - 1}{\theta_f}} \right)^{\frac{\theta_f}{\theta_f - 1}} \tag{3.2}$$

$$F(s) = E \cdot \frac{\Gamma(s) P(s)^{1-\theta_f}}{\sum_{s' \in \mathcal{S}} \Gamma(s') P(s')^{1-\theta_f}}, \qquad P = \left(\sum_{s \in \mathcal{S}} \Gamma(s) P(s)^{1-\theta_f} \right)^{\frac{1}{1-\theta_f}} \tag{3.3}$$

[4] Appendix A describes the entry block in detail: the timing of entry, the entrant cohort, and how operating and sunk costs are identified.

[5] The wage is the numeraire ($w = 1$, labor supplied perfectly elastically), and the final expenditure level, the cell demand levels $A(k)$, and the baseline normalization $P_V^0(s) = 1$ are level parameters that rescale nominal magnitudes without affecting any allocation, elasticity or counterfactual reported below. They are omitted in what follows.

Across-sector optimal allocation $F(s)$ is a share of the exogenous expenditure level E. Equation 3.3 shows the optimality condition for the optimal expenditure allocation in sector s. $P(s)$ denotes the sectoral price index while P denotes the economy-wide price index.

3.2 Within-cell demand

Within each cell s, an exogenous number of firms $\mathcal{N}_s$ produce the cell good $Y(s)$. Each firm is denoted by the subindex i, and its production is denoted by $y_i(s)$. The elasticity of substitution across firms within sector s is denoted by γ and is assumed to be common across all cells s. Similar to the aggregate case, the firm-level market shares $\sigma_i(s)$ depend on firm-level prices and the within-sector elasticity of substitution γ. Finally, the quantity produced per firm is denoted by $q_i(s)$, and it depends on the total expenditure in sector s denoted by $E(s)$

$$Y(s) = \left(\sum_{i \in \mathcal{N}_s} y_i(s)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \qquad \gamma > \theta_f \tag{3.4}$$

$$\sigma_i(s) = \frac{p_i(s)^{1-\gamma}}{\sum_{j \in \mathcal{N}_s} p_j(s)^{1-\gamma}}, \qquad P(s) = \left(\sum_{i \in \mathcal{N}_s} p_i(s)^{1-\gamma} \right)^{\frac{1}{1-\gamma}} \tag{3.5}$$

$$q_i(s) = \sigma_i(s) \frac{E(s)}{p_i(s)} \tag{3.6}$$

4 Technology

I assume each firm uses three inputs for production: labor, denoted by $L_i(s)$; materials, denoted by $X_i(s)$; and capital, $K_i(s)$. Materials are a CES bundle across cells with elasticity κ. The share of cell s's materials bill spent on goods from cell s' is denoted by $\varpi_{ss'}$. Crucially, all firms within a cell s face the same materials input prices $P^x(s)$, that depends on the sectoral price indexes of their upstream sectors.

$$X_i(s) = \left(\sum_{s'} \varpi_{ss'}^{\frac{1}{\kappa}} x_i(s,s')^{\frac{\kappa-1}{\kappa}} \right)^{\frac{\kappa}{\kappa-1}}, \tag{4.1}$$

$$P^x(s) = \left(\sum_{s'} \varpi_{ss'} P(s')^{1-\kappa} \right)^{\frac{1}{1-\kappa}} \tag{4.2}$$

$$W_{ss'} = \frac{\varpi_{ss'} P(s')^{1-\kappa}}{\sum_{s''} \varpi_{ss''} P(s'')^{1-\kappa}} \tag{4.3}$$

4.1 Variable bundle

I assume firms produce by combining a CES aggregator of labor and materials, with a sector-specific elasticity of substitution $\eta(s)$, and capital. $\alpha(s)$ denotes a sector-specific CES share parameter that captures the relative importance of labor in production. $P_v(s)$ denotes the sector level bundle price index, and w denotes wages. Finally, I denote $\phi_L(s)$ as the sector-level labor cost share of the variable bundle.

$$V_i(s) = \left(\alpha(s) L_i(s)^{\frac{\eta(s)-1}{\eta(s)}} + \big(1-\alpha(s)\big) X_i(s)^{\frac{\eta(s)-1}{\eta(s)}} \right)^{\frac{\eta(s)}{\eta(s)-1}} \tag{4.4}$$

$$P_V(s) = \left(\alpha(s)^{\eta(s)} w^{1-\eta(s)} + \big(1-\alpha(s)\big)^{\eta(s)} P^x(s)^{1-\eta(s)} \right)^{\frac{1}{1-\eta(s)}} \tag{4.5}$$

$$\phi_L(s) = \frac{\alpha(s)^{\eta(s)} w^{1-\eta(s)}}{P_V(s)^{1-\eta(s)}} \tag{4.6}$$

To generate output, firm i combines capital and the variable bundle in a Cobb-Douglas production function, with capital elasticity of output of α_K. Firm-level productivity is denoted by $z_i(s)$.

$$y_i(s) = z_i(s) K_i(s)^{\alpha_K} V_i(s)^{1-\alpha_K} \tag{4.7}$$

4.2 Cost and the Capital Restriction

Firms in reserved cells face a constrained problem, given the statutory restriction on Plant and Machinery capital. I assume that Plant and Machinery capital, denoted by $B_i(s)$ is a fraction of

productive capital $K_i(s)$. The conversion rate is given exogenously by $\tau_i(s)$, which is heterogeneous across firms within sector s. Intuitively, this parameter is intended to capture different production composition in terms of capital. It rationalizes observed firms with high market shares and high productivity that report low levels of Plant and Machinery capital. Formally, this is described by equation 4.8.

$$B_i(s) = \frac{K_i(s)}{\tau_i(s)} \tag{4.8}$$

The statutory restriction implies that, in reserved sectors, firms may use up to $\overline{B}$ units of Plant and Machinery capital. This maps into a firm-level restriction of total capital as follows:

$$\overline{K}_i(s) = \tau_i(s)\overline{B} \tag{4.9}$$

4.3 Unconstrained problem

Firms in never-reserved sectors face an unconstrained problem, which yields expressions for marginal costs and optimal capital demand as in equations 4.10 and 4.11.

$$\lambda_i^*(s) = \frac{1}{z_i(s)} \left(\frac{r}{\alpha_K}\right)^{\alpha_K} \left(\frac{P_V(s)}{1-\alpha_K}\right)^{1-\alpha_K} \tag{4.10}$$

$$K_i^*(s) = \frac{\alpha_K \lambda_i^*(s) q_i(s)}{r} \tag{4.11}$$

4.4 Constrained problem

For $K_i^*(s) > \bar{K}_i(s)$ the constraint binds, V_i. Hence, constrained firms solve (4.7) at $K_i = \bar{K}_i(s)$, implying a variable bundle described by equation 4.12

$$V_i(s) = \left(\frac{q_i(s)}{z_i(s)\bar{K}_i(s)^{\alpha_K}}\right)^{\frac{1}{1-\alpha_K}} \tag{4.12}$$

Firm i total costs are given by equation 4.13, and the implied marginal cost curve is 4.14.

$$\text{TC}_i(s) = r\bar{K}_i(s) + P_V(s)V_i(s) \tag{4.13}$$

$$\lambda_i(s) = \frac{P_V(s)}{1-\alpha_K}\left(\frac{1}{z_i(s)\bar{K}_i(s)^{\alpha_K}}\right)^{\frac{1}{1-\alpha_K}} q_i(s)^{\frac{\alpha_K}{1-\alpha_K}} \tag{4.14}$$

so that, collecting (4.10) and (4.14), it is clear that capped firms operate at higher marginal costs. The magnitude of the shift is given by the gap between the optimal capital and the cap.

$$\lambda_i(s) = \begin{cases} \lambda_i^*(s) & K_i^*(s) \leq \bar{K}_i(s) \\ \lambda_i^*(s)\left(\dfrac{K_i^*(s)}{\bar{K}_i(s)}\right)^{\frac{\alpha_K}{1-\alpha_K}} & K_i^*(s) > \bar{K}_i(s) \end{cases} \tag{4.15}$$

5 Pricing

I assume that firms compete à la Bertrand. Markups are endogenously determined and vary with market share. I assume that $\gamma > 1, \text{and}, \theta_f > 1$, and $\gamma > \theta_f$. This means firms are more substitutable within markets than across sectors in aggregate demand. I denote the perceived elasticity of substitution $\varepsilon_i(s)$ under the nested CES structure, which depends only on γ, θ_f, and the market shares, by equation 5.1. In equilibrium, elasticities map to markups as in equation D.4

$$\varepsilon_i(s) = \gamma\big(1-\sigma_i(s)\big) + \theta_f\sigma_i(s) \tag{5.1}$$

$$\mu_i(s) = \frac{\varepsilon_i(s)}{\varepsilon_i(s)-1}, \qquad p_i(s) = \mu_i(s)\lambda_i(s) \tag{5.2}$$

5.1 Entry

5.1.1 Operating Costs

Incumbents' participation is rationalized by a sectoral-level operating cost $f^R(s)$. The operating cost rationalizes the observed number of surviving incumbents recovered from the data. I assume that, in the baseline, incumbents pay only the operating cost and have already paid the sunk discovery cost in a period the model does not consider.

Under the previous assumptions, the operating recurring cost in the reserved and never reserved lines cells must not exceed the gross profit of the least profitable incumbent:

$$f^R(s) = \min_{i \in \mathcal{I}_R(s)} \Pi_i^0(s), \qquad f^U(s) = \min_{i \in \mathcal{I}_U(s)} \Pi_i^0(s), \tag{5.3}$$

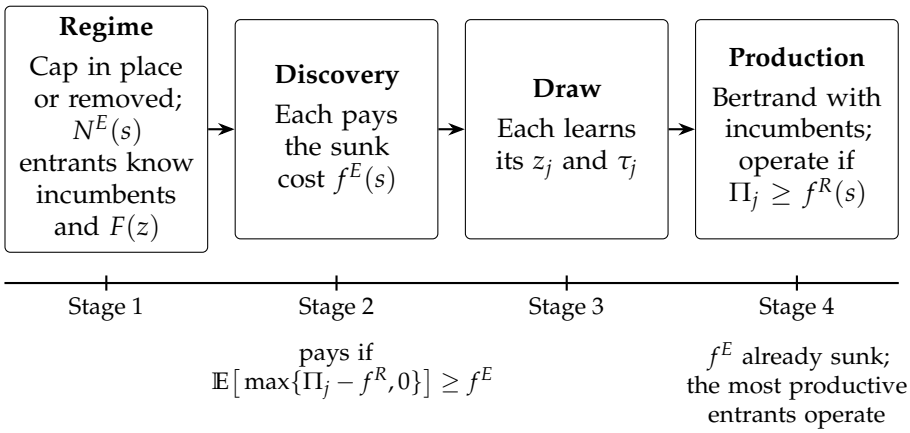
where $\Pi_i^0(s)$ is incumbent i's gross profit in the baseline equilibrium, and $\mathcal{I}_R(s)$ and $\mathcal{I}_U(s)$ are the observed incumbents in the reserved and never-reserved cells, respectively. I select the highest operating cost consistent with all observed incumbents producing; any lower value would also be consistent with survival. Choosing the upper bound makes the operating stage as demanding as the data allow, so entry after the reform is not driven by an arbitrarily low operating cost.

5.1.2 Sunk Costs

Firms' entry is rationalized by a sector-specific discovery sunk cost denoted by $f^E(s)$. Ex-ante identical potential entrants pay the sunk cost to learn their productivity draw. I assume the number of entrants is predetermined and equal to the median number of observed entrants in the three years following the reform. The argument that follows the identification of $f^E(s)$ is a revealed-preference argument. I ask which $f^E(s)$ rationalizes two conditions evaluated in expectation. First, that for the pre-determined pool of entrants it is not worthwhile to enter in the baseline, where the policy was in place. Second, once the cap is removed, observed entry is rationalized, and entrants can cover sunk and operational costs with uncapped profits.

Entry requires the operating surplus to cover the recurring cost $f^R(s)$ and the annualised sunk cost $f^E(s)$. Two bounds are calculated: An upper bound assuming no cap in place and a lower

Figure 4: Timing of entry



bound with a capped economy.

$$\underbrace{\frac{1}{N^E(s)} \sum_{j \in \mathcal{J}(s)} \max\left\{\Pi_j^{cc} - f^R(s),\, 0\right\}}_{\substack{\text{Expected option value} \\ \text{Cap in place}}} \;<\; f^E(s) \;\leq\; \underbrace{\frac{1}{N^E(s)} \sum_{j \in \mathcal{J}(s)} \max\left\{\Pi_j^{ff} - f^R(s),\, 0\right\}}_{\substack{\text{Expected option value} \\ \text{Cap removed}}} \tag{5.4}$$

and $f^E(s)$ is set identified on $\left[\underline{f}^E(s), \overline{f}^E(s)\right]$, with ψ selecting a point:

$$f^E(s) = \underline{f}^E(s) + \psi\left(\overline{f}^E(s) - \underline{f}^E(s)\right) \mathbf{1}\left\{\overline{f}^E(s) > \underline{f}^E(s)\right\}, \qquad \psi = \tfrac{1}{2} \tag{5.5}$$

6 Labor Demand and Factor Shares

Labor is recovered from variable expenditure net of capital rental, which holds in both regimes of (4.15):

$$\ell_i(s) = \phi_L(s) \frac{P_V(s) V_i(s)}{w} = \phi_L(s) \frac{\mathrm{TC}_i(s) - rK_i(s)}{w} \tag{6.1}$$

$$\mathrm{TC}_i(s) - rK_i(s) = \begin{cases} (1 - \alpha_K) \lambda_i^*(s) q_i(s) & K_i^*(s) \leq \bar{K}_i(s) \\ P_V(s) \left(\dfrac{q_i(s)}{z_i(s) \bar{K}_i(s)^{\alpha_K}} \right)^{\frac{1}{1-\alpha_K}} & K_i^*(s) > \bar{K}_i(s) \end{cases} \tag{6.2}$$

$$L(s) = \sum_{i \in \mathcal{N}_s} \ell_i(s), \qquad \mathcal{V}(s) = \sum_{i \in \mathcal{N}_s} \big(\mathrm{TC}_i(s) - rK_i(s)\big) \tag{6.3}$$

The cell-level revenue shares that close the network are

$$\varphi_L(s) = \frac{\phi_L(s)\mathcal{V}(s)}{E(s)}, \qquad \varphi_M(s) = \frac{\big(1-\phi_L(s)\big)\mathcal{V}(s)}{E(s)}, \qquad \varphi_K(s) = \frac{r\sum_{i \in \mathcal{N}_s} K_i(s)}{E(s)} \tag{6.4}$$

6.1 Decomposition of labor demand

Because every firm's variable cost equals $(1-\alpha_K)$ times its marginal cost times output, in both regimes of (4.15), the variable-cost share of a cell's revenue depends on markups only:

$$v(s) \equiv \frac{\mathcal{V}(s)}{E(s)} = \varphi_L(s) + \varphi_M(s) = \frac{1-\alpha_K}{\bar{\mu}(s)}, \qquad \bar{\mu}(s) = \Big(\sum_{i \in \mathcal{N}_s} \sigma_i(s)\mu_i(s)^{-1}\Big)^{-1}, \tag{6.5}$$

where $\bar{\mu}(s)$ is the revenue-weighted harmonic mean of firm markups. Labor employment therefore factors as

$$L(s) = \frac{\varphi_L(s)E(s)}{w} = (1-\alpha_K) \times \underbrace{\frac{E(s)}{w}}\ \text{scale} \times \underbrace{\frac{1}{\bar{\mu}(s)}}\ \text{markup} \times \underbrace{\phi_L(s)}\ \text{complementarity}, \tag{6.6}$$

From (4.5)–(4.6), labor's share of the variable bundle can be written as

$$\phi_L(s) = \left[1 + \left(\frac{1-\alpha(s)}{\alpha(s)}\right)^{\eta(s)} \left(\frac{P^x(s)}{w}\right)^{1-\eta(s)}\right]^{-1}, \tag{6.7}$$

so it depends on prices only through the relative price of materials, raised to $1-\eta(s)$. Let $\Pi^x(s)$ denote the change in that relative price between the two equilibria. The complementarity term is then exactly

$$\mathcal{C}(s) = -\log\Big[\phi_L^0(s) + \big(1-\phi_L^0(s)\big)\Pi^x(s)^{1-\eta(s)}\Big], \qquad \Pi^x(s) = \frac{P^{x,1}(s)/P^{x,0}(s)}{w^1/w^0}, \tag{6.8}$$

which depends only on the elasticity $\eta(s)$, the baseline labour share $\phi_L^0(s)$ and the change in the

relative price of materials. When materials become cheaper ($\Pi^x(s) < 1$):

$$\mathcal{C}(s) \begin{cases} > 0 & \eta(s) < 1 \text{ (complements: the bracket is below one)} \\ = 0 & \eta(s) = 1 \text{ (Cobb–Douglas)} \\ < 0 & \eta(s) > 1 \text{ (substitutes: the bracket is above one)} \end{cases} \tag{6.9}$$

and its size grows with $|1 - \eta(s)|$. To first order,

$$\begin{aligned} \mathcal{C}(s) &\approx \big(1 - \eta(s)\big)\big(1 - \phi_L^0(s)\big) g(s), \\ g(s) &= -\log \Pi^x(s) \approx \sum_{s'} W_{ss'} \big(-\Delta \log P(s') \big), \end{aligned} \tag{6.10}$$

The Expression 6.10 summarizes the effects on labor demand that propagate through the network and depend on the elasticity of substitution between labor and intermediate inputs.

At this point, I can summarize the expected effects of a policy that induces competition and price changes using equation 6.6. In my framework, policy works by favoring entry and incumbent expansion. Entry directly affects market shares within each affected market, subsequently affecting markup according to expression 5.1. In equilibrium, the policy affects the pricing condition and the sectoral price indexes $P(s)$. The magnitude of the shock propagates as a function of the input-output linkages $W_{ss'}$ and the changes in sectoral prices. The complementarity term $\mathcal{C}(s)$ depends on the sign of the elasticity $\eta(s)$.

7 Market Clearing

$$\mathbf{R} = \Big(\mathbf{I} - \mathbf{W}^\top \operatorname{diag}\big(\varphi_M(s)\big) \Big)^{-1} \mathbf{F} \tag{7.1}$$

with $\mathbf{F}$ collecting (3.3), and cell expenditure $E(s) = R(s)$.

$$w = 1, \qquad E = \bar{E}, \qquad L = \sum_{s \in \mathcal{S}} L(s), \quad L(s) = \sum_{i \in \mathcal{N}_s} \ell_i(s) \tag{7.2}$$

Labor is supplied perfectly elastically at the numeraire wage, so aggregate employment L is

determined by labor demand (6.1) and is not fixed; net jobs created are drawn from outside the modelled sectors. Nominal final expenditure is held at its baseline level. Counterfactual employment changes are changes in labor demand at a given wage and nominal spending

8 Parametrization

Productivity Distribution.— I assume that across sectors the number of incumbent firms is predetermined and set equal to the average number of incumbents in the pre-policy period. Within a sector s, a single Pareto distribution with shape parameter $\xi(s)$ governs the productivity $z_i(s)$ of the entire pool of incumbent producers. Firms are ordered by $z_i(s)$ and assigned by rank: the most productive occupy the never-reserved cell, the next block the reserved cell, and the remainder stay inactive. $\xi(s)$ is identified from the dispersion of market shares and the HHI within the never-reserved cell.

I assume that reserved production additionally requires a product-specific capability denoted by $a_i(s)$. A firm assigned to the reserved cell produces with effective productivity $z_i(s)a_i(s)$, where

$$a_i(s) = \exp\big(\sigma(s)\max\{q_i, 0\}\big), \qquad q_i = \Phi^{-1}\Big(\tfrac{i-1/2}{n}\Big),$$

and q_i is the firm's rank-quantile of the standard normal. The single parameter $\sigma(s)$ is chosen to match the observed market-share quantiles and the Herfindahl index of the reserved cell.

Finally, I assume that potential entrants in deregulated sectors draw productivity from a Pareto distribution that is partially informed by the average incumbent productivity $\bar{z}_s^R$:

$$z_{js}^E = \bar{z}_s^R \varphi_s (1 - u_{js})^{\frac{-1}{\alpha_s^E}} \tag{8.1}$$

Where φ_s is the sector-specific entrant location, α_s^E is the entrant Pareto tail parameter, and u_{js} is a deterministic quantile grid; I calibrate the location parameter to match the total entrants' sales share and the tail parameter to match the entrant conditional HHI. The support is exogenously imposed as the median observed number of entrants in the 3 next periods after the reform. In the sectors that do not have enough entrants to identify a sector-specific shape, I use a pooled shape parameter calculated as the mean of the Pareto shapes calculated identified from suitable sectors.

Fixed costs.— The parametrization of sunk costs follows the next logic. As described in section 5.1, the incumbent predetermined number of firms are not making entry decisions. Hence, sunk costs apply only to potential entrants. I recover the sectoral-level sunk costs following a revealed-preference argument. Intuitively, I use the sectoral sunk-cost-free parameter to rationalize potential entrants' decision not to enter in the baseline period, when the capital restriction is in place. Given that I assume that the number of incumbents in each cell comes from the data, the operating cost is rationalized by the gross profit of the least profitable incumbent. Sunk costs are then identified from actual exogenously imposed entrant and incumbent counts. Importantly, this argument rationalizes productive entrants' decision not to enter while the cap is active, as restricted profits do not compensate for the extra sunk cost they must pay compared to a similarly productive incumbent.

The operating-cost rationale follows similar logic. I choose the highest operating cost consistent with the observed incumbents continuing production with positive profits. This identification argument selects the upper bound, and any lower sector-level operating cost would also imply incumbent survival. Operating costs explain why incumbents participate in the market, while sunk costs explain why potential entrants stay out.

Capital.— I assume that there is a firm-level map from productive capital to plant and machinery capital. The parameter $\tau_i(s)$ captures how many productive capital services are obtained from a unit of plant and machinery. This parameter captures either firm-level heterogeneity in machinery prices, capital quality or capital composition. I assume $\tau_i(s)$ follows a log-normal distribution. The location parameter identifies the median plant and machinery capital for entrants, while the dispersion identifies the fraction of firms with plant and machinery capital above the statutory restriction. For incumbents, $\tau_i(s)$ disciplines the number of firms in the [8M 10M] plant and machinery bin. I select this bin to align with reduced-form evidence showing that expansion is concentrated in this bin.

Externally Calibrated Parameters.— I impose a set of parameters following the standard values used in the literature. I set the elasticity of substitution γ within sectors to 6, following Edmond et al. (2023), and $\theta_f = 1.25$ across sectors, following Edmond et al. (2015). I impose an elasticity of substitution between inputs sourced within the firm $\kappa = 2.5$, following long-run across materials

elasticities found by Peter and Ruane (2025). I also externally calculate the Input-Output $\omega_{ss'}$ weights using firm-level data on 3-digit sector-level input consumption. I use single-product data to isolate the specific consumption of output classified in each 3-digit sector, a common method for isolating unobserved input-allocation effects in product-level datasets (Rachapalli, 2021; Orr, 2022; De Loecker et al., 2016). Using single-product firms exclusively allows me to classify consumption properly, without confounding potential unobserved input sharing within firms across different 3-digit sectors.

Finally, I externally estimate the sectoral-level elasticity of substitution between labor and materials, as described in section 1.3. I use those externally estimated elasticities in the equilibrium model calculations[6]. This is the main departure from the literature and the heterogeneity that drives the labor-materials complementarity channel.

9 Calibration Results

9.1 Parameter Estimates

I present the main parameters calibrated in the model in table 12. I find relatively thin tails in the productivity distribution of never-reserved sectors (Pareto shape parameter = 4.3). Operating costs in never-reserved sectors are about 90 times higher than in reserved sectors, while sunk costs are about 300 times larger than the recurring operating costs in reserved sectors. The distribution that describes entrants exhibit fatter tails than incumbents', with a shape parameter of 3.42.

9.2 Calibration Validation

To validate the model, in this section I present two main results. First, in table 3[7] I show the general fit of my model to the targeted moments, in reserved and never reserved sectors. Generally, The

[6]Given that the sectoral elasticities are estimated noisily, I present the main results of this section in Appendix F, using the shrunk estimates. I shrink each estimated η towards its mean depending on how large is the sectoral level parameter estimate standard error. There are not dramatic changes in the main results

[7]The number of sectors in Table 3 is smaller than the number of sectors the counterfactual de-regulates. The counterfactual removes the cap in 113 reserved cells. The table reports each moment only on the sectors where it can be compared between data and model. In Panel A, a sector is considered if it has at least four incumbents. This keeps 79 of the 113 reserved cells and 207 of the 236 never-reserved cells. In Panel B, the sample is the 84 reserved sectors with documented post-reform entrants in the data. For those with no entrants, in the calibration, their entry targets are pooled: the entrant sales share and capital moments are the medians across the sectors with documented entrants, and the number of entrants is the median entry rate times the sector's number of incumbents.

model reproduces fairly well the concentration distribution in the economy. In particular, the model captures well not only the level values for the market shares, but also captures correctly the sectoral distribution, as Pearson correlation coefficient across sectors is close generally higher than 75%.

Panel A in table 3 show the fit for incumbents in the baseline period. Here, the model does a better work capturing the moments of reserved sectors, while overstating the Never-Reserved sectors concentration. In general, the concentration moments that I find for India are close to the ones reported for Taiwan by Edmond et al. (2015) for Taiwan. In Panel B, I report the moments associated to entrants. The model captures correctly the entrant count and their sales shares. Also, The capital implied by the model seems to reproduce the policy effects. I find that the percent of entrants above cap in my model is around 35% while in the data is about 33%. Similarly, conditional on being above the cap, entrants enter with capital four times higher than the cap in the data, and about 5 times in the model.

I validate the main entry and expansion mechanisms. In appendix C, I show that upon cap removal, entry happens in firms that otherwise, during the capped phase could not cover their operating and sunk discovery cost, due to be operating in the increasing marginal cost section of the curve. Once the cap is removed, the firms fully optimize without the additional restriction and the revenues overcome the fixed costs (Figure 6). Also, I show that incumbents expansion is concentrated in previously constrained incumbents whose capital used was exactly at the cap. Moreover, the expansion percentage in the aggregate fairly reproduces the Differences-in-Differences model at the firm level (Figure 7).

Table 3: Model fit: incumbent market structure and post-reform entry

Market	Moment	Sectors	Data	Model	Pearson
Panel A. Baseline incumbent market structure					
Reserved	HHI	79	0.151	0.139	0.943
	Market-share p25 (percent)	79	0.29	0.28	0.888
	Market-share p50 (percent)	79	0.96	0.98	0.995
	Market-share p75 (percent)	79	2.71	2.62	0.973
Never reserved	HHI	207	0.120	0.167	0.873
	Market-share p25 (percent)	207	0.16	0.28	0.916
	Market-share p50 (percent)	207	0.67	0.64	0.762
	Market-share p75 (percent)	207	1.85	1.70	0.931
Panel B. Post-reform entrants					
Entrants	Entrant count	84	12.61	12.69	1.000
	Entrant sales share (percent)	84	19.88	18.48	0.814
	P&M p50 / cap	83	0.288	0.390	–
	P&M above cap (percent)	83	33.142	35.366	–
	P&M p50 / cap, conditional on above cap	83	4.148	5.432	–

Notes: The unit of observation is a three-digit sector. Panel A compares survey-weighted incumbent moments with the calibrated baseline. It uses sectors with at least four incumbents and strictly ordered p25, p50, and p75 shares in both data and model, giving a common sample within each market type. Market shares and their means are reported in percent. The never-reserved moments are calibration targets. For reserved markets, HHI, p50, and p75 are targeted; p25 provides a common distributional diagnostic (the calibration also uses p90, not shown). Panel B uses the median across the four post-reform snapshots T through $T+3$. Entrant counts are survey-expanded in the data and rounded when imposed on the model. Entrant sales share and sector-level median plant and machinery and the sector share above the Rs. 10 million cap are targeted. The three capital rows are pooled distributional summaries over the common 83-sector sample: data pool survey-weighted entrant-period observations from T through $T+3$, while the model pools final-equilibrium entrants with equal weight. Plant and machinery is the entrant establishment's total capital; within-firm product shares recover that total and the table does not report product-line capital. The conditional p50 is an untargeted upper-tail diagnostic. Model moments use the final network equilibrium. One data-supported sector has no final-equilibrium entrant; it is included as zero for counts and sales and omitted from the capital-distribution rows.

10 Counterfactual: Removing $\overline{K}$

Deregulating sector k removes the restriction from $R(k)$, with incumbent expansion and entrants unrestricted. Next, I show the policy's implications for key parameters. My main object of interest is labor demand. However, before presenting the results on the key object, I will show how the model operates on prices and productivity. Then, I decompose employment gains into a scale component, a markup component, and the substitution channel.

10.1 Market Outcomes: Prices, markups and productivity

10.1.1 Prices and Markups

Before presenting the main results, this section shows the main market outcomes affected by the policy and the intermediate channels through which the policy transmits to labor demand. I start by showing the aggregate effects of cap removal on output and materials prices. The first noteworthy result is that entry drives most of the total effect on output and materials prices. This result is consistent with Martin et al. (2017) findings, which show that entrants drive most of the positive outcomes generated by dereservation policy in India. In particular, the entry of more productive firms into deregulated sectors is associated with a 5.24% reduction in the market output price index, holding expansion constant. This contrasts with the solo-expansion effect, which accounts for a -0.34% reduction in the output price index of reserved sectors.

The reduction in output prices propagates through the network, driving down material prices not only in deregulated sectors but also in never-reserved sectors. I find a 0.42% reduction in material prices in never-reserved sectors through propagation. Because of variable markups, the reduction does not fully pass through to output prices (-0.35%). The price propagation mechanism is central to the substitution channel outside the deregulated sectors. In a model without labor-materials complementarities, labor-demand effects in never-reserved sectors arise only through demand reallocation. When labor-materials complementarities are considered, the price propagation affects the labor decision through substitution between labor and materials.

Finally, markup decline is concentrated in deregulated sectors, as the model structure allows entry only in this set of sectors. Because firms pay homogeneous sectoral-level material prices, relative prices are not distorted in the never-reserved sectors after the pro-competition policy. This explains the lack of markup movement in these sectors. Despite this, the effect is not exactly zero, as I do observe firm exits in general equilibrium, although their magnitude is too small to meaningfully move markups.

10.1.2 Productivity

Now I move to evaluate the policy effects on aggregate productivity. I follow Edmond et al. (2015) and calculate aggregate productivity under two scenarios: with the equilibrium allocation of output

Table 4: Effects of de-reservation on prices and markups, by margin

		Expansion	Entry	Total
All sectors	Output price	−0.04	−0.66	−0.69
	Materials price	−0.02	−0.43	−0.45
	Markup	0.00	−0.03	−0.03
Reserved sectors	Output price	−0.34	−5.24	−5.52
	Materials price	−0.04	−0.75	−0.78
	Markup	−0.01	−0.48	−0.48
Never-reserved sectors	Output price	−0.01	−0.34	−0.35
	Materials price	−0.02	−0.41	−0.42
	Markup	0.00	0.00	0.00

Notes: Entries are percentage changes (100× the log change) from the pre-reform equilibrium, averaged within each group with baseline revenue weights. *Expansion* removes the cap on incumbents (up to the data-implied expansion ceiling) and does not allow entry; *Entry* admits the entrant cohort and holds incumbents at the statutory cap; *Total* removes the cap for incumbents and admits entrants together, as in the main counterfactual. Each column is a separate network equilibrium. The two margins interact through cell demand, so Expansion and Entry need not sum to Total. The materials price is the input bundle aggregated over suppliers through the input–output network, so it moves only through network prices; the markup is $\varepsilon/(\varepsilon-1)$ with $\varepsilon = \gamma(1-s)+\theta s$, so it moves only where market shares move. Entrants: 1,210 in the entry run, 0 in the expansion run.

across firms, which is distorted by markup dispersion and, before the reform, by the capital cap; and with the efficient allocation that a planner would choose given the same set of firms and productivities. For each sector s, productivity under the equilibrium allocation is

$$a(s) = \Big(\sum_i \big(\mu_i/\mu(s)\big)^{-\gamma} a_i^{\gamma-1} \Big)^{\frac{1}{\gamma-1}}, \tag{10.1}$$

Under the efficient allocation, all firms charge the same markup, and sector productivity becomes $a^e(s) = \big(\sum_i a_i^{\gamma-1}\big)^{\frac{1}{\gamma-1}}$. The ratio $a(s)/a^e(s) \leq 1$ measures allocative efficiency. I decompose the change in $a(s)$ into changes in the efficient benchmark (productivity and the number of active firms) and in allocative efficiency.

Table 5 shows that removing the capital cap raises productivity in deregulated sectors by 4.1 log points. Most of the effect comes through the entry of firms that are more productive than the typical incumbent. Moreover, allocative efficiency rises by 0.24 log points, indicating lower markup dispersion and a general shift of output production toward more productive firms.

Table 5: Change in aggregate productivity after de-reservation

	Total	Efficient	Allocative
De-regulated lines	+4.11	+3.87	+0.24
Never-reserved sectors	0.00	0.00	0.00
Economy	+0.27	+0.25	+0.02

Notes: Change in sector productivity $a(s)$, 100× the log change. A sector s is a three-digit industry × reservation status. Following Edmond, Midrigan and Xu (2015), $a(s) = \left(\sum_i (\mu_i/\mu(s))^{-\gamma} a_i^{\gamma-1}\right)^{1/(\gamma-1)}$, where a_i is firm i's productivity, $\mu_i = p_i/(W/a_i)$ its markup over the unit cost of the composite input, $\mu(s) = p(s)/(W/a(s))$ the sector markup, and $\gamma = 6$. For a firm held at the capital cap, marginal cost exceeds W/a_i, so the cap enters through μ_i. Equivalently, $a(s)$ is sector output per unit of composite input. *Efficient* is the change in the first-best benchmark $\left(\sum_i a_i^{\gamma-1}\right)^{1/(\gamma-1)}$, which reflects firm productivities and the number of firms; *Allocative* is the change in $a(s)$ relative to that benchmark, from markup dispersion and the capital cap. The two sum to the total. Sector changes are aggregated with pre-reform revenue weights; entrants are included after the reform.

10.2 Employment Gains

Now, I measure the employment gains from the pro-competitive policy by attributing them to each channel, using 100 jobs created in deregulated lines as the base measure.[8] The change reflects a shift from the baseline capped equilibrium to an equilibrium where $\overline{K}$ is removed, and entrants and incumbents can enter the market freely, regardless of capital level. I split the employment gains by deregulated and never-reserved sectors. Within each cell, I split according to the elasticity of substitution η.

As reported in Table 6, for every 100 jobs created in deregulated lines, 57 jobs survive. The substitution channel explains about a third of the total net job creation $(\frac{+20.1}{+56.9})$. Three features deserve special attention, as they are the main mechanisms in my model. First, heterogeneity in labor and material elasticity of substitution cushions the effect in never-regulated sectors. This is because, given the distribution of η, price propagation through the network affects the labor demand positively, and larger, more connected sectors are classified as complements. Naturally, this is an empirical feature in the Indian economy, although it is consistent with recent evidence pointing to general complementarity between labor and materials across economies.

A second noteworthy feature is that most of the aggregate cushion effect operates through the network. Particularly, around 83% of the total effect of the substitution channel is generated by sectors that were never touched by the policy. Again, this is a natural result, as the policy

[8] Appendix B collects the labor-demand and factor-share equations on which the decomposition is based.

targeted sectors that represented about 10% of the employment and 6% of the revenue in the economy. However, this result shows that, on top of the input-output linkages, complementarity heterogeneity is an important mechanism when evaluating labor effects of industrial policies.

Finally, a striking result is that the substitution channel in the aggregate is more important than the traditional pro-competitive channel. This is an anticipated feature of small and targeted industrial pro-competitive policy, as the one studied in this paper. This is an important result because the competition literature has focused on the distortion reduction implied by trade liberalization or industrial policy, and the potential employment benefits from this (Peters, 2020; Liu, 2019). The novel result presented here draws attention to an alternative mechanism that is shaped by the sectoral-level heterogeneous technology and generates non-negligible effects on labor outcomes. This points to an additional dimension that policymakers must consider when designing industrial policy that drives job creation.

Moving to the other mechanisms, around half of the effect in the aggregate is explained by the scale component ($\frac{26.7}{56.9}$). Intuitively, this is driven by the reallocation that happens across sectors through three channels. First, final consumers substitute across sectors with an elasticity $\theta > 1$. Pro-competitive policies that promote entry negatively affect sectoral-level price indexes. Hence, final demand reallocates towards the set of directly regulated sectors. Second, in my model I allow an elastic substitution across inputs bought within the firm, following Peter and Ruane (2025). Downstream firms that use deregulated products as inputs also shift their purchases toward the now-cheaper suppliers. Finally, cheaper inputs also lower the deregulated lines' own costs, which amplifies the demand shift through the network.

Around 18% of the effect is explained by the markup effect ($\frac{+10.1}{56.9}$). The markup effect captures the pro-competitive effects of entry. Most of the action in this component happens in the deregulated lines, as the model generates entry incentives exclusively in this set of sectors, consistent with the real policy environment. Entry reshapes market shares and affects markup distribution. Overall, my findings show the expected pro-competitive effects: markups decline due to increased competition, implying reduced sector market distortions. As the markup distribution shifts down, labor demand increases, which helps explain the effects we find.

In never-reserved lines, the effect is negative but very close to zero. Because this set of sectors

was not subject to the policy shift, the entry effect is null. Moreover, because of the structure of our input-price index, we do not allow differential changes in firms' marginal costs due to the cost shock originating in upstream affected markets. Hence, this channel does not affect markups as it holds relative prices constant. The only mechanism in the model that affects markups in the never-reserved sectors is firms' potential exit due to reduced profits as sector revenues decline from reallocation. Although present, the magnitude of this mechanism is negligible.

Table 6: Decomposition of the employment effect of de-reservation, per 100 jobs created in de-regulated lines

	Scale	Markup	Complementarity	Total
De-regulated lines				
$\eta < 1$	+69.1	+8.1	+3.6	+80.8
$\eta > 1$	+17.4	+1.9	−0.2	+19.2
All	+86.5	+10.1	+3.4	+100.0
Never-reserved sectors				
$\eta < 1$	−46.1	0.0	+19.2	−26.9
$\eta > 1$	−13.7	0.0	−2.5	−16.2
All	−59.8	0.0	+16.7	−43.1
Net, economy-wide	+26.7	+10.1	+20.1	+56.9

Notes: Each entry is the change in employment attributed to a channel and group, per 100 jobs created in de-regulated lines (the de-regulated rows' total). Cells are three-digit sector × reservation status; de-regulated lines are the reserved cells of the 113 treated sectors, and never-reserved sectors include all other cells. The decomposition uses economy-wide logarithmic-mean (Sato–Vartia) weights, so rows are additive and the net row is the sum of the two group totals. Scale is the change in log revenue; Markup is minus the change in the log markup, measured as the revenue-weighted harmonic mean of firm markups in the cell, which equals the change in the log share of revenue spent on labour and materials; Complementarity is the change in the log labour share of variable cost. With the wage as numeraire the three sum to the change in log employment. η is the sector's labour–materials elasticity of substitution. The de-regulated lines' employment effect is 0.355 percent of baseline economy-wide employment (about 25,321 jobs at survey-expanded baseline employment); the economy-wide effect is 0.202 percent.

10.3 Role of IO Network

As discussed above, the Input-Output network in my model plays an important role by propagating material prices, amplifying the scale effect, and serving as the channel through which complementarity operates. In this section, I ask how the total net effects change when we turn off the input price propagation mechanism. In Table 7, I present the results of this exercise. Three things deserve attention. First, the Input-Output propagation triples the net employment effect ($\frac{56.9}{19.2} = 2.96$). This amplification is in line with Input-Output multipliers found for economies similar to India in terms of the materials share of revenue, as in China (Jones, 2013). Second, it is

important to highlight that, under my model structure, the markup channel is not sensitive to the Input-Output structure assumption, as it is driven mainly by the entry effects induced by the policy. Finally, of the 38 extra jobs that Input-Output propagation adds, 53% are created by the substitution channel. Without complementarity, cheaper materials would only raise output. With it, they also change how much labor each unit of output requires. Therefore, the network determines which sectors are indirectly affected by the policy, but the elasticities of the sectors it reaches determine the sign and magnitude of the effect, which is why heterogeneity in η matters for the aggregate employment response.

Table 7: Employment effects with and without price transmission through the network, per 100 jobs created in de-regulated lines in the baseline

	Scale	Markup	Complementarity	Total
Baseline (full network)				
De-regulated lines	+86.5	+10.1	+3.4	+100.0
Never-reserved sectors	−59.8	0.0	+16.7	−43.1
Net, economy-wide	+26.7	+10.1	+20.1	+56.9
Network prices fixed				
De-regulated lines	+16.4	+9.4	0.0	+25.9
Never-reserved sectors	−6.7	0.0	0.0	−6.7
Net, economy-wide	+9.8	+9.4	0.0	+19.2

Notes: Every block is scaled per 100 jobs created in de-regulated lines in the baseline, so the blocks compare levels; construction as in Table 6. The baseline solves the full network equilibrium. With network prices fixed, material and input-bundle prices are held at their pre-reform values and input–output shares are fixed, so neither price changes nor supplier substitution are transmitted through the network; revenue linkages through the input–output system remain active. With input prices fixed the labour share of variable cost does not move, so the complementarity component is zero.

10.4 Role of the Elasticity of Substitution level and heterogeneity

I am now ready to quantify the extent to which heterogeneity in labor-material substitution affects the labor outcome measured by the model. To do this, I recalibrate the model under three additional scenarios. First, I impose a homogeneous elasticity of substitution equal to 0.7. The second scenario corresponds to my pooled estimate of equation 1.1, which is 0.76. This captures the economy-wide elasticity of substitution at the cost of shutting down heterogeneity. Finally, I consider a Cobb–Douglas specification, which imposes a unit elasticity of substitution. I report the results in table 8.

Compared to the baseline, the results show a non-negligible discrepancy in the policy-induced

net jobs gain relative to the heterogeneous benchmark. In particular, in the third row of Table 8, I show that using the homogeneous pooled elasticity of 0.76 leads to an estimated job creation about 14% larger than the benchmark. In this scenario, we remove all the negative effects generated by the sectors that substitute labor and materials, and overestimate the effects of all sectors that, although complements, stand in the [0.76-1] elasticity bin. This overestimation compensates for the underestimation driven by all the sectors that exhibit elasticities lower than 0.76. A similar logic applies to collapsing to the Cobb-Douglas case, although the conclusion is the opposite: The estimation of job creation is about 30% lower compared to the benchmark case.

Another important implication of the results exposed above is that policymakers who design industrial policy might be looking at the wrong object when trying to foster employment in the economy. In column 4 of table 8, I calculate the aggregate real output gain generated by the policy. As expected, output gains are not sensitive to the elasticity of substitution between labor and materials. This is informative about the limited ability to translate output gains into labor gains without a consistent estimate of η.

Table 8: Complementarity experiments: employment and output

	Employment, per 100 jobs created in de-regulated lines		Output gain (%)
	Net	Complementarity	
Baseline (estimated η)	+56.9	+20.1	+0.71
$\eta = 0.70$	+70.9	+36.2	+0.70
$\eta = 0.76$	+64.9	+29.3	+0.71
$\eta = 1$	+39.7	0.0	+0.71

Notes: Economy-wide employment effect (Net) and its complementarity component per 100 jobs created in de-regulated lines, each specification normalised by its own de-regulated effect. The baseline uses the estimated sector-specific labour–materials elasticities η; the other rows impose a common η on every sector and re-solve the network equilibrium with all other parameters unchanged. With $\eta = 1$ the labour share of variable cost is invariant to input prices, so the complementarity component is zero. Output gain is the economy-wide change in real gross output, $100\times$ the log change of each market's output (revenue deflated by its price index) aggregated with pre-reform revenue weights. The mean of the estimated η weights each cell's elasticity by its baseline employment.

To investigate further the specific role of *heterogeneity* in the elasticity of substitution η, I build an index that captures the extent to which each sector s is connected to complementary or substitute sectors.

$$\begin{aligned} \text{Compl}(s) &= \Big[\boldsymbol{\mathcal{L}}\left(\boldsymbol{\vartheta}\odot(\mathbf{1}-\boldsymbol{\eta})\right)\Big]_s = \sum_{s'}\mathcal{L}_{ss'}\,\vartheta(s')\left(1-\eta(s')\right), \\ \boldsymbol{\mathcal{L}} &= \left(\mathbf{I}-\mathbf{W}^{\top}\,\text{diag}\left(\varphi_M(s)\right)\right)^{-1}, \end{aligned} \tag{10.2}$$

where $W_{s's}$ is the share of buyer s''s materials spending on supplier s in (4.3), evaluated at baseline prices, $\varphi_M(s')$ is the materials share of revenue in (6.4), so that $\boldsymbol{\mathcal{L}}$ is the inverse in the revenue system (7.1), $\vartheta(s') = E(s') / \sum_{s''} E(s'')$ are revenue weights, and $\eta(s')$ is the labor–materials elasticity of substitution. $\mathcal{L}_{ss'}$ measures the direct and indirect pass-through of a fall in s's price to s''s materials price, and the factor $\big(1 - \eta(s')\big)$ signs the labor-share response, $\partial \log \phi_L(s') / \partial \log P^x(s') = -\big(1 - \eta(s')\big)\big(1 - \phi_L(s')\big)$. A positive index means that price reductions in s reach mostly sectors where labor and materials are complements; a negative index, mostly substitutes.

The idea is to show that comparable industrial policies that generate similar output effects and target similar sectors can differ substantively in their job-creation predictions when heterogeneity is considered. To show this, I simulate two counterfactual policies that affect 8 different sectors that share similar characteristics. The only substantive difference is the effective elasticity of substitution between labor and materials for downstream firms, given by the average $\text{Compl}(s)$ of the targeted sectors.

Table 9 summarizes the key results of this exercise. In column 2, I present the policy effects when the policy is intentionally built to reach sectors that are complements according to the $\text{Compl}(s)$ index, and in column 3, I show the effects when the reached sectors are substitutes. Using the estimated technologies, I show a non-negligible gap of 13 jobs between the number of net jobs created by the two policies. The one reaching complements creates 44% more jobs than the one reaching substitutes. If the same policies were evaluated assuming a common elasticity of substitution, the gap is of only 3 jobs (Only 6%), and a policymaker would see these reforms as roughly equivalent. This reinforces my argument that evaluating the entire technology distribution in the economy is essential to adequately assess the policy's labor effects. Furthermore, this exercise illustrates why output effects alone are insufficient to anticipate labor outcomes when elasticity is heterogeneous.

Finally, note that the reform that reaches complements generates about 25% *fewer* jobs when evaluated using the estimated technologies than in the common-elasticity case. The reform reaching substitutes creates about 45% *fewer* jobs than the common elasticity case. This points to within-policy divergence in effects, not only across policies.

I empirically generalize this result in figure 5. I generate a set of 49 random policy pairs that

differ only in terms of the gap in the effective elasticity of buyers reached. In the chart, it is clear that as the gap of reached elasticity grows favoring substitute sectors reached, the gap between the policies job creation becomes more negative. Using a common elasticity, the gap moves around zero as expected. In particular, when one policy price cuts reach buyers whose effective elasticity is 0.25 higher (more substitutes) than its matched counterpart's, it creates about 1 fewer net jobs per 100 workers employed in the de-regulated sectors.

The conclusion of this section is that the homogeneous elasticity of substitution assumption frequently used in the literature leads to a significant mismeasurement of the effects of the policy in terms of job creation, even though the output effects are correctly measured. If policymakers care about job creation when implementing industrial policies, it is crucial to understand not only the Input-Output linkages in the economy, but also the technology of sectors affected through propagation effects.

Figure 5: Elasticity Gap of Reached Sectors vs Jobs Creation

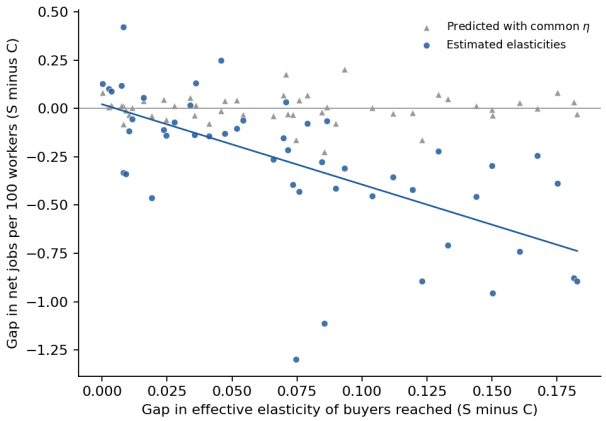


Table 9: Two comparable reforms with different employment effects

	Reaches complements	Reaches substitutes
Panel A. Comparable characteristics		
Employment de-regulated (% of total)	0.71	0.71
Output gain (basis points)	4.2	4.3
Labor share of revenue (%)	7.3	6.7
Materials share of revenue (%)	66.6	66.5
Average markup	1.22	1.23
Own labor–materials elasticity (average)	0.92	0.92
Jobs created in de-regulated sectors, per 100 workers	2.4	2.5
Panel B. Employment effects with estimated technologies		
Effective elasticity of the buyers reached	0.86	0.97
Net jobs, per 100 workers in de-regulated sectors	1.1	0.8
Net jobs, per 100 created in de-regulated sectors	43.3	30.1
Jobs displaced elsewhere, per 100 created	56.7	69.9
Labor–materials channel, per 100 created	22.3	3.2
Panel C. Prediction with a homogeneous elasticity ($\eta = 0.76$)		
Net jobs, per 100 workers in de-regulated sectors	1.4	1.4
Net jobs, per 100 created in de-regulated sectors	58.2	55.0
Jobs displaced elsewhere, per 100 created	41.8	45.0
Labor–materials channel, per 100 created	38.0	30.1

Notes: Each reform removes the reservation cap in a set of reserved sectors (complements: 122, 214, 291, 415, 424, 653, 661, 742; substitutes: 323, 361, 414, 425, 553, 571, 750, 763); the remaining reserved sectors keep the cap. The two packages have the same average own elasticity and are matched on employment de-regulated, output gains, and the net jobs and jobs created that a common elasticity $\eta = 0.76$ predicts (single-sector de-regulations); among such pairs, they are the one whose price cuts reach buyers with the most different elasticities. Panels A and B use the elasticities estimated at the two-digit level (one-digit where a two-digit estimate is unavailable), assigned to every three-digit sector. Shares and markups are revenue-weighted pre-reform averages over the de-regulated cells; the own elasticity is the simple average over the package's sectors. The effective elasticity weights each cell's elasticity by its employment, its materials share of labor and materials costs, and the fall in its materials price. Jobs displaced elsewhere are jobs created in de-regulated sectors minus net jobs economy-wide. Panel C solves the same two reforms with a common elasticity $\eta = 0.76$ imposed on every sector; its per-100 rows use the jobs that model creates in the de-regulated sectors.

11 Conclusion

In this paper, I study how industrial policy propagates through production networks in economies with variable markups and heterogeneous elasticity of substitution between labor and materials. I show that the labor-for-materials substitution elasticity is a key parameter for correctly evaluating the effects of industrial policy. Propagation through this channel has a larger effect on aggregate job creation than the pro-competitive effects traditionally studied through markups. Importantly, I show that policymakers interested in promoting job creation through small targeted industrial policies must take into account the elasticity between labor and materials of indirectly affected sectors, as it has substantive effects on the outcome that cannot be predicted with first-order output

propagation effects.

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A Entry: Operating Costs, Sunk Costs and Identification

This appendix describes how entry into reserved product lines is modelled and how its parameters are identified. Entry is the margin through which most of the price decline in de-regulated lines arises, so the assumptions behind it matter for the counterfactuals. The block has three components: recurring operating costs, which every producer pays; a cohort of potential entrants, whose size and productivity distribution are disciplined by the data; and a sunk discovery cost, which rationalises why these entrants stay out while the cap is in place and enter once it is removed.

A.1 Timing

Entry unfolds in two stages. In the *discovery stage*, a potential entrant pays a sunk cost $f^E(s)$ before learning its productivity in the reserved line. In the *operating stage*, it observes its productivity and the capital ceiling it faces, and operates if its gross profit covers the recurring operating cost $f^R(s)$. Because $f^E(s)$ is paid before productivity is known, the entry decision is taken in expectation over the productivity distribution of the cohort, and $f^E(s)$ is sunk when the operating decision is made. Incumbents have already paid it and face only $f^R(s)$.

A.2 Operating costs

Operating costs are identified from the survival of incumbents. Every observed incumbent operates in the baseline, so the recurring cost in the reserved cell must not exceed the gross profit of the least profitable incumbent:

$$f^R(s) = \min_{i \in \mathcal{I}_R(s)} \Pi_i^0(s), \qquad f^U(s) = \min_{i \in \mathcal{I}_U(s)} \Pi_i^0(s), \tag{A.1}$$

where $\Pi_i^0(s)$ is incumbent i's gross profit in the baseline equilibrium and $\mathcal{I}_R(s)$, $\mathcal{I}_U(s)$ are the observed incumbents of the reserved and never-reserved cells. I select the largest operating cost consistent with all observed incumbents producing; any lower value would also be consistent with survival. Choosing the upper bound makes the operating stage as demanding as the data allow, so entry after the reform is not driven by an arbitrarily low operating cost.

A.3 The entrant cohort

Number of entrants.— The number of entrants $N^E(s)$ in each reserved sector is taken from the data: the survey-expanded count of entrants into the reserved product lines, measured as the median over the reform period and the three subsequent periods (T to $T+3$) and rounded to an integer. Taking the median over four periods smooths the timing of entry, which is spread over several years after de-reservation. Of the 113 reserved sectors, 84 have complete four-period entrant moments; the remaining 29 use the common-tail calibration described below. Entry is therefore not free: the model takes the observed number of entrants as given and asks what productivity distribution and costs are consistent with it.

Productivity.— Entrants draw productivity from a Pareto distribution anchored to incumbents, evaluated on a deterministic quantile grid:

$$z^E_{js} = \bar{z}^R_s \, \varphi_s \, (1 - u_{js})^{-\frac{1}{\alpha^E_s}}, \qquad u_{js} = \frac{j - \frac{1}{2}}{N^E(s)}, \quad j = 1, \ldots, N^E(s), \tag{A.2}$$

where $\bar{z}^R_s$ is the mean productivity of the sector's reserved incumbents, φ_s a location and α^E_s a tail parameter. The location φ_s is chosen to match the sales share of entrants in the reserved product lines after the reform: given the incumbents, a higher location makes entrants larger. The tail α^E_s is chosen to match the concentration of sales among entrants, measured by their Herfindahl index: a thinner tail spreads sales more evenly across entrants. The target Herfindahl index weights equally the observed four-period index and the index implied by the pooled tail, which limits the influence of sectors with few entrants; where the observed index lies below its feasible minimum $1/N^E(s)$, it is set to that minimum.

The tail is identified sector by sector in 75 of the 113 reserved sectors. Elsewhere I use a pooled tail,

$$\alpha^E = (\theta_f - 1) + \exp\Big(\tfrac{1}{|\mathcal{A}|} \textstyle\sum_{s \in \mathcal{A}} \log \left(\alpha^E_s - (\theta_f - 1) \right) \Big) = 3.42, \tag{A.3}$$

where $\mathcal{A}$ is the set of 72 sectors whose tail is identified in the interior of its admissible range. The admissible range requires $\alpha^E_s > \theta_f - 1$, so that expected sales are finite.

Capital.— Entrants convert productive capital into plant and machinery at a firm-specific rate

τ_{js}, drawn from a log-normal distribution with location μ_s^τ and dispersion σ_s^τ. The location matches the median plant-and-machinery capital of entrants relative to the cap, and the dispersion matches the share of entrants above the cap. The draws are ordered so that the most productive entrants hold the most plant and machinery per unit of productive capital, consistent with the evidence that entrants into de-reserved lines were large, capital-intensive producers. The conversion rate does not affect prices or allocations once the cap is removed. It matters only for reported capital and for the value of entering under the cap, because the statutory ceiling applies to plant and machinery.

A.4 Sunk costs

The sunk cost is identified by revealed preference from the timing of entry. The cohort did not enter while the cap was in place but entered once it was removed. Define two values for each entrant j:

- Π_j^{cc}, its gross profit if it entered alone under the cap, facing the capped incumbents at pre-reform prices and expenditure;
- Π_j^{ff}, its gross profit in the post-reform equilibrium, together with the whole cohort of $N^E(s)$ entrants and with incumbents released to $\chi(k)\bar{K}$, where $\chi(k)$ is sector k's data-implied expansion ceiling.

Because the sunk cost is paid before productivity is known, an entrant compares it with the expected value of the option to operate, averaged over the cohort's productivity distribution. No entry under the cap and entry after its removal then bound $f^E(s)$:

$$\underline{f}^E(s) \;<\; f^E(s) \;\leq\; \overline{f}^E(s), \qquad \begin{aligned} \underline{f}^E(s) &= \frac{1}{N^E(s)} \sum_{j=1}^{N^E(s)} \max\left\{\Pi_j^{cc} - f^R(s),\, 0\right\}, \\ \overline{f}^E(s) &= \frac{1}{N^E(s)} \sum_{j=1}^{N^E(s)} \max\left\{\Pi_j^{ff} - f^R(s),\, 0\right\}. \end{aligned} \tag{A.4}$$

The lower bound is the expected option value of entering under the cap; the upper bound, the expected option value once the cap is removed. The maximum reflects the option not to operate after a poor draw. The removal of the cap rationalises entry when $\overline{f}^E(s) > \underline{f}^E(s)$, and $f^E(s)$ is then

set identified on that interval. I select its midpoint,

$$f^E(s) = \underline{f}^E(s) + \tfrac{1}{2}\big(\overline{f}^E(s) - \underline{f}^E(s)\big). \tag{A.5}$$

The interval is non-empty in 89 of the 113 reserved sectors. In the remaining 24, the observed cohort earns less in expectation after the reform than the option value of entering under the cap. No sunk cost rationalises both margins there, and I record $f^E(s) = \underline{f}^E(s)$.

A.5 Operating stage in equilibrium

In the counterfactual equilibrium the cohort has already paid $f^E(s)$. Entrants are ordered by productivity and operate as long as each covers its operating cost, given the equilibrium created by those that operate:

$$n^*(s) = \max\Big\{n \le N^E(s) \;:\; \min_{j \le n} \Pi_j^{ff}(n) - f^R(s) \ge 0\Big\}. \tag{A.6}$$

Charging $f^E(s)$ at this stage would count it twice. Of the 1,223 entrants in the cohort, 1,210 operate in equilibrium, in 104 of the reserved cells.

A.6 Summary of identification

Table 11 summarises how each parameter of the entry block is pinned down. Three features of the strategy are worth stressing.

First, the extensive margin is taken from the data rather than generated by the model. The number of entrants is the observed count, and their productivity distribution is disciplined by two moments of the observed entrants: their aggregate sales share, which fixes how large entrants are relative to incumbents, and the concentration of their sales, which fixes how unequal they are among themselves. Their capital is disciplined by the median capital of entrants relative to the cap and the share above it. The counterfactual therefore reproduces how many firms entered and how large they were; what it adds is the general-equilibrium response of the rest of the economy.

Second, the cost parameters are identified by revealed preference. The operating cost is the largest value consistent with every observed incumbent producing. The sunk cost is bounded by

the requirement that the same cohort, valued in expectation before its productivity is known, finds entry unprofitable under the cap and profitable without it. In the 89 sectors where these bounds are consistent, the cap is the only change between the two values, so the timing of entry identifies the range of the sunk cost; the midpoint is a normalisation within that range.

Third, given the fixed entrant count, the sunk cost does not affect the counterfactual allocation: it is sunk at the operating stage. Its role is to verify that the calibrated cohort is consistent with the observed timing of entry, rather than to determine the scale of entry. What determines the employment effects of entry is the size and productivity of the entrants, which are tied to observed entrant moments, and the operating cost, which is tied to incumbent survival. The entry moments are targeted; the responses of incumbents near the cap are not, and serve as external validation of the model (Table 10).

Table 10: Model fit: entrant and incumbent margins

Margin	Moment	N	Data	Model	Pearson	Spearman
Panel A. Post-reform entrants (targeted)						
Extensive	Entrants per sector	84	12.61	12.69	1.000	0.989
Intensive	Entrant sales share (percent)	84	19.88	18.48	0.814	0.929
	P&M p50 / cap	83	0.288	0.390	–	–
	P&M above cap (percent)	83	33.14	35.37	–	–
Panel B. Incumbents, by pre-reform plant and machinery (untargeted)						
Intensive	Δ log P&M, Rs. 8–10m	120	0.178 (0.085)	0.071	–	–
	Difference, Rs. 8–10m minus below Rs. 8m	–	0.200	0.208	–	–
Extensive	Above Rs. 10m after reform, from Rs. 8–10m	120	0.558	0.613	–	–
	Remaining in Rs. 8–10m, from Rs. 8–10m	120	0.157	0.268	–	–

Notes: Panel A repeats the entrant rows of Table 3: the unit of observation is a three-digit sector, data use the median over post-reform snapshots T through $T+3$, and model moments use the final network equilibrium. Entrant counts are imposed; entrant sales share and the sector-level median plant and machinery and above-cap share are calibration targets, while the capital rows report pooled entrant distributions. Panel B is not targeted. Incumbents are plants that made a reserved product before de-reservation; bins use pre-reform plant and machinery, and the Rs. 8–10m bin includes plants at the Rs. 10m cap. Data Δ log P&M are the bin coefficients of the Figure 2 fixed-effects regression (standard error in parentheses). Model values are the change in log total plant and machinery of incumbents active before and after the reform, averaged within bin. The difference row subtracts from the Rs. 8–10m value the average over the five bins below Rs. 8m (bin coefficients in the data, bin means in the model). Model incumbents also lose share to entrants, which lowers every bin by a common amount; the difference removes that level. Extensive-margin rows use survivors observed before and after de-reservation, weighted by survey multipliers, and classify each plant by its mean post-reform plant and machinery. N is the number of data plants (Panel B); the corresponding model incumbents are 697 in Rs. 8–10m and 10,214 below Rs. 8m.

Table 11: Identification of the entry block

Parameter	Identified from	Source
Operating cost $f^R(s)$, $f^U(s)$	Survival of the least profitable incumbent	Baseline equilibrium, observed incumbents
Number of entrants $N^E(s)$	Observed count, median over T to $T+3$	Survey-expanded plant data
Productivity location φ_s	Sales share of entrants in reserved lines	Post-reform plant data
Productivity tail α_s^E	Herfindahl index of entrant sales (pooled tail where not identified)	Post-reform plant data
Capital conversion μ_s^τ, σ_s^τ	Median entrant capital relative to the cap; share of entrants above the cap	Post-reform plant data
Sunk cost $f^E(s)$	Expected option values under the cap and without it, eq. (A.4); midpoint	Model, given the cohort

B Labor Demand and Factor Shares

Labor is recovered from variable expenditure net of capital rental, which holds in both regimes of (4.15):

$$\ell_i(s) = \phi_L(s)\,\frac{P_V(s)V_i(s)}{w} = \phi_L(s)\,\frac{\mathrm{TC}_i(s) - rK_i(s)}{w} \tag{B.1}$$

$$\mathrm{TC}_i(s) - rK_i(s) = \begin{cases} (1-\alpha_K)\,\lambda_i^*(s)\,q_i(s) & K_i^*(s) \le \bar{K}_i \\ P_V(s)\left(\dfrac{q_i(s)}{z_i(s)\bar{K}_i^{\alpha_K}}\right)^{\frac{1}{1-\alpha_K}} & K_i^*(s) > \bar{K}_i \end{cases} \tag{B.2}$$

$$L(s) = \sum_{i\in\mathcal{N}_s} \ell_i(s), \qquad \mathcal{V}(s) = \sum_{i\in\mathcal{N}_s} \big(\mathrm{TC}_i(s) - rK_i(s)\big) \tag{B.3}$$

The cell-level revenue shares that close the network are

$$\varphi_L(s) = \frac{\phi_L(s)\mathcal{V}(s)}{E(s)}, \qquad \varphi_M(s) = \frac{\big(1-\phi_L(s)\big)\mathcal{V}(s)}{E(s)}, \qquad \varphi_K(s) = \frac{r\sum_{i\in\mathcal{N}_s} K_i(s)}{E(s)} \tag{B.4}$$

C Validation

Figure 6: Entry Validation

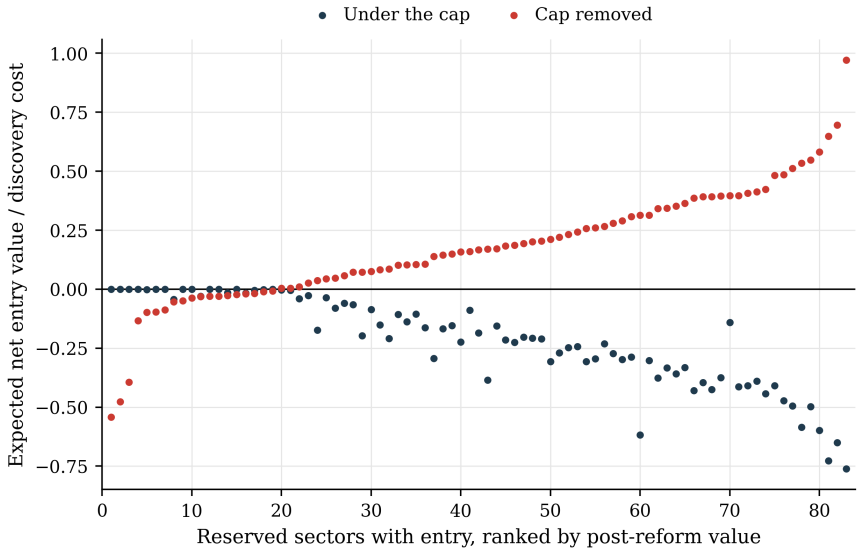


Figure 7: Expansion validation

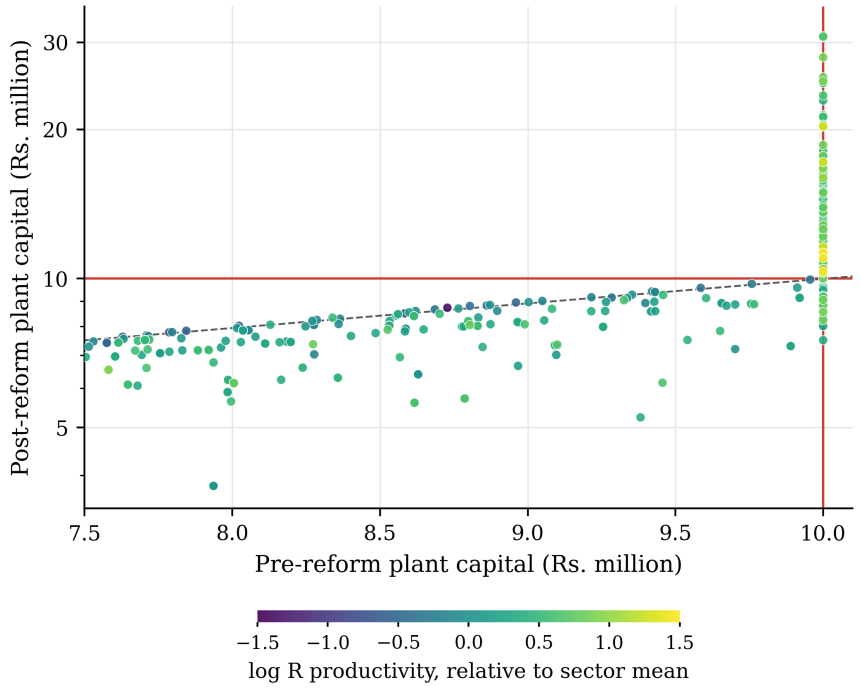


Table 12: Parameter estimates

Parameter	Description	Source / target	Value	Dispersion
Panel A. Fixed or externally estimated				
γ	Within-market elasticity	Set	6	—
θ_f	Across-market elasticity	Set	1.25	—
κ	Supplier substitution elasticity	Literature	2.5	—
α_K	Capital share	Set	0.10	—
r	Rental rate	Set	0.10	—
$\overline{B}$	Statutory cap on plant and machinery	Policy	Rs. 10m	—
$\eta(s)$	Labor–materials elasticity	Two-digit sector estimates	med. 0.90	range [0.36, 1.73]
Panel B. Incumbents (per sector)				
$\xi(s)$	Productivity Pareto shape, all sectors ($n = 238$)	U share quantiles, HHI_U	4.30	IQR [3.16, 5.44]
$\xi(s)$	sectors with a reserved branch ($n = 113$)	same	4.58	IQR [3.47, 5.56]
$\sigma(s)$	Dispersion of reserved-type advantage	R share quantiles, HHI_R	0.336	IQR [0.258, 0.412]
$\alpha(s)$	Labor weight in variable bundle	Factor shares	0.085	IQR [0.054, 0.128]
$f^R(s)$	Operating cost, reserved	Marginal incumbent's gross profit	1.1×10^{-6}	IQR [2.5×10^{-7}, 1.5×10^{-5}]
$f^U(s)$	Operating cost, never-reserved	Marginal incumbent's gross profit	9.7×10^{-5}	IQR [2.8×10^{-5}, 4.9×10^{-4}]
$\tau_i(s)$	Capital conversion rate, eligible reserved firms ($n = 108$)	Eligible plant and machinery	3.68	IQR [1.75, 7.30]; p90 20.1
$K_i^*/\overline{K}_i$	Cap severity, firms at the cap	Target 1.20	1.07	IQR [1.02, 1.39]
Panel C. Entrants (per sector)				
α_s^E	Entrant productivity Pareto shape (75 sectors identified)	Entrant HHI; pooled 3.42 elsewhere	3.42	IQR [3.29, 3.62]
φ_s	Entrant Pareto minimum / mean incumbent productivity	Entrant sales share	0.86	IQR [0.74, 0.99]
τ_{js}	Entrant capital conversion rate ($n = 1210$ entrants)	Median P&M / cap, share above cap	0.68	IQR [0.14, 2.10]; firms p10–p90 [0.03, 5.3]
$f^E(s)$	Discovery (sunk) cost ($n = 83$)	Midpoint of [pre, post] expected value	1.9×10^{-4}	IQR [7.4×10^{-5}, 6.6×10^{-4}]
f^E/f^R	Sunk vs. operating cost	Implied	302	IQR [46, 1,517]
$N^E(s)$	Entrants per sector	Data (four-period median)	5	IQR [1, 12]
Panel D. Margins of adjustment in the counterfactual				
	Entrants operating	Model	1,210	of 1,223 in the cohort
	Reserved sectors with entry	Model	104	of 113 reserved sectors
	Eligible incumbents increasing plant and machinery	Model	429	4.1% of 10,443
	of which ending above Rs. 10m	Model	427	of 519 at the cap before
	Reserved sectors with incumbent expansion	Model	57	of 108 with eligible incumbents

Notes: The incumbent productivity grid is common to reserved and never-reserved incumbents of a sector; reserved incumbents' productivity is that grid times a type advantage with dispersion $\sigma(s)$. Operating costs are the gross profit of the marginal incumbent. Cap severity is desired capital at baseline output, under the firm's own $\tau_i(s)$, relative to observed total plant and machinery, among eligible firms at the cap (pooled). Entrant productivity is Pareto with a

D Markup Estimation

D.1 Theoretical Problem

Consider a production function for a firm f producing a product j at time t

$$Q_{fjt} = F(L_{fjt}, M_{fjt}, K_{fjt})\Omega_{ft} \tag{D.1}$$

Where Q_{fjt} is physical output, L_{fjt} and M_{fjt} are labour and materials measures (that usually are assumed to be a variable input that the firm can freely adjust), K_{fjt} is a capital measure that is assumed to be a fixed input that faces adjustment costs. Denote the W^x_{fjt} as the variable input v price. Finally, Ω_{ft} is the TFP (In this case, TFPQ, as I estimate a quantity-based production function).

Assuming that plants minimize short-run costs, taking output quantity and input prices as given, the first order condition associated with the Lagrangian function $\mathcal{L}$ with respect to materials for any product and time is:

$$\frac{\partial \mathcal{L}_{fjt}}{\partial M_{fjt}} = W^m_{fjt} - \lambda_{fjt}\frac{\partial Q_{fjt}}{\partial M_{fjt}} = 0 \tag{D.2}$$

A crucial assumption here is that costs are separable across products, which allows for product-by-product cost minimization. In equation D.2, λ_{fjt} denotes the marginal cost of production at a given output level. One can rearrange equation D.2 as follows:

$$\frac{\partial Q_{fjt}}{\partial M_{fjt}}\frac{M_{fjt}}{Q_{fjt}} = \frac{1}{\lambda_{fjt}}\frac{W^m_{fjt}M_{fjt}}{Q_fjt} \tag{D.3}$$

The left-hand side in equation D.3 represents the output-elasticity with respect to materials, which I denote as θ. Moreover, since markup (μ_{fjt}) is defined as the ratio between prices and marginal costs $\left(\frac{P_{fjt}}{\lambda_{fjt}}\right)$, equation D.3 can be rewritten as follows:

$$\mu_{fjt} = \theta_{fjt} \left(\frac{P_{fjt} Q_{fjt}}{W^m_{fjt} M_{fjt}} \right) \tag{D.4}$$

Notice that using ASI, the expression $\left(P_{fjt} Q_{fjt}\right)$ is observed at the product level. However, $\left(W^m_{fjt} M_{fjt}\right)$ is observed only at the plant level. Hence, two challenges remain to estimate consistent expressions for TFPQ and marginal costs: first, estimate θ and second, obtain expressions to map the plant-level input expenditures to a product-level counterpart.

D.2 Estimation

To estimate the production function described above, consider first a log version adding a measurement error and unanticipated shocks to output denoted by ϵ_{fjt}

$$q_{jft} = f\left(l_{fjt}, m_{fjt}, k_{fjt}; \mathbf{fi}\right) + \omega_{ft} + \epsilon_{fjt} \tag{D.5}$$

Lower case variables represent the log of the variables described in equation D.1.

Following De Loecker et al. (2016), I estimate the production function exploiting data for multi-product firms that record revenue and quantities sold at the product level. The objective is to obtain consistent estimates of f, which implies the vector β estimation that determines the output elasticities with respect to the physical inputs.

Employing a quantity-based production function mitigates concerns regarding the output-price bias described above. However, potential biases arise when input prices and product-level input allocations are not observable, which is often the case with production databases such as the ASI. This implies that the product-level inputs, l_{fjt}, m_{fjt} and k_{fjt}, are typically unobservable and that the vector $\left(\hat{l}_{ft}, \hat{m}_{ft}, \hat{k}_{ft}\right)$ of firm-level deflated input expenditures is observed instead.

To understand the issue, consider the relationship between product-level input quantities and firm-level for any variable input expenditures denoted by x,

$$x_{fjt} = \rho_{x_{fjt}} + \hat{x_{ft}} - \hat{w}_{x_{fjt}}, \tag{D.6}$$

where $\rho_{x_{fjt}}$ is the (log) share of input expenditures attributable to product j and $\hat{w}_{x_{fjt}}$ denotes the deviation of the unobserved (log) firm–product-specific variable input price from the (log) industry-wide variable input price index used to deflate input expenditures. The same principle applies to each component of the deflated input expenditures vector. Consequently, (D.5) can be written as:

$$q_j = f\left(l_{ft}, m_{ft}, k_{ft}\right) + A\left(\rho_{fjt}, l_{ft}, m_{ft}, k_{ft}\right) + B\left(w_{fjt}, \rho_{fjt}, l_{ft}, m_{ft}, k_{ft}\right) + +\omega_{ft} + \epsilon_{fjt}, \tag{D.7}$$

Where functions A and B are unobservable due to the absence of information about ρ_j and p_j, the vector of (log) input expenditure shares and price deviations by product. The bias emerges from the correlation between the unobservable productivity ω and A and B via the deflated input expenditures.

To address the bias coming from unobserved product-level inputs, De Loecker et al. (2016) rely on single-product firms, for whom $\rho_{x_{fjt}} = 0$. Thus, the bias for these firms comes only from function B. To mitigate this bias, the authors argue that high-quality products produced using high-quality inputs command higher final prices. This theoretical connection establishes a positive relationship between unobserved product-specific input prices and observed product-specific output prices. I employed a second-degree polynomial to approximate the relation between input and output prices. After controlling this, I estimated the production function assuming a trans-log production function using the instrumental variables approach proposed by Wooldridge (2009).

I can retrieve the output elasticity related to the intermediate input θ_m and ρ_m using the results. Following De Loecker and Warzynski (2012), I can recover markups as follows

$$\mu_{fjt} = \theta_m \cdot \frac{P_{fjt} Q_{fjt}}{e^{\rho_{fjt}} W_{ft}^M M_{ft}} \tag{D.8}$$

Where $e^{\rho_{fjt}}$ is the share of input expenditures attributable to the product j. ρ_{fjt} is retrieved by solving a within-plant system of equations. To execute the estimation algorithm, I use the deflated. intermediate inputs as m, the deflated wage bill as l and k is calculated using the perpetual inventory method.

E Instrumental Variables for the Elasticity of Substitution

Table 13: Pooled Elasticity of Substitution: Distance-Based Spatial Instruments

Specification	η	s.e.	KP F	Hansen p
OLS	0.914	0.006	–	–
IV, Z^{200} (main)	0.758	0.128	31.0	–
IV, $Z^{w,200}$ and $Z^{M,200}$, 2SLS	0.771	0.143	33.1	0.83
Robustness of the main IV				
Adding log capital	0.738	0.139	26.0	–
Sector + 2-digit × year + state effects	0.810	0.089	63.7	–
Sector × year effects, no state effects	0.900	0.106	55.9	–
Sector × year + district effects	0.807	0.116	31.1	–

Notes: 2SLS estimates of equation 1.1 with 3-digit sector-by-year and state effects unless stated otherwise; controls are firm age and type of organization. Standard errors are clustered by district. KP F is the Kleibergen–Paap Wald statistic. 92,609 firm-years (88,172 with the local labor-market wage).

At the sector level, however, these instruments are weak. Across the 40 two-digit sectors, the median first-stage F is 2.3 and only three sectors have $F \geq 10$. The instruments vary only at the district-sector-year level, and within a single sector there is too little of that variation left after the fixed effects are absorbed. I therefore use the design below for the sector elasticities.

E.1 Sector elasticities: long differences and an input-price shift-share instrument

Following Peter and Ruane (2025), I estimate equation 1.1 in long differences between a base year t_0 and $t_1 = t_0 + h$, for horizons $h = 1, \ldots, 4$. Let Δ denote the change between t_0 and t_1. The estimating equation is

$$\Delta \log \left(\frac{P_i^M M_i}{w_i L_i} \right) = (\eta_s - 1) \, \Delta \log \left(\frac{w_i}{P_i^M} \right) + \delta_{h,t_0} + \lambda_{h,n(i)} + \varepsilon_i, \tag{E.1}$$

where δ_{h,t_0} are horizon-by-base-year effects, $\lambda_{h,n(i)}$ are horizon-by-sector effects for the firm's base-year 3-digit sector $n(i)$, and the differencing removes time-invariant differences in technology and in the units in which firms report inputs.

Input prices. Let p_{kjt} be the unit value of input j reported by firm k in year t, and $\mathcal{B}_{jt}^{-d}$ the set of firms that buy input j in year t outside district d. For each input, unit of measurement, and year,

the national input price is the leave-own-district-out mean log unit value,

$$\bar{p}_{jt}^{-d} = \frac{1}{N_{jt}^{-d}} \sum_{k \in \mathcal{B}_{jt}^{-d}} \log p_{kjt}.$$

Before averaging, I correct unit-of-measurement errors and drop item-year cells in which the national median jumps by more than one log point and firm-level unit values more than $\log 20$ away from the item mean. Let $s_{ij,t}$ be the share of input j in firm i's domestic materials expenditure in year t. The change in the firm's material price is the Törnqvist index of national input-price changes,

$$\Delta \log P_i^M = \frac{\sum_j \bar{s}_{ij} \, \Delta \bar{p}_j^{-d(i)}}{\sum_j \bar{s}_{ij}}, \qquad \bar{s}_{ij} = \tfrac{1}{2} \left(s_{ij,t_0} + s_{ij,t_1} \right).$$

Instrument. The instrument fixes the firm's input basket at its base-year shares (a Laspeyres index):

$$Z_i = \frac{\sum_j s_{ij,t_0} \, \Delta \bar{p}_j^{-d(i)}}{\sum_j s_{ij,t_0}}. \tag{E.2}$$

Z_i differs from $\Delta \log P_i^M$ only in that it uses base-year rather than average shares, so it does not respond to the firm's own input choices after t_0. It is a shift-share instrument whose shifts are national input-price changes driven by demand and supply conditions outside the firm's district, and whose exposure shares are the firm's own base-year basket. Because the basket differs across firms in the same sector, the instrument varies firm by firm and remains strong within sectors. Exclusion requires that national input-price changes for the items in the firm's base basket be uncorrelated with changes in the firm's factor-bias term and with its own measurement error, conditional on the horizon-by-base-year and horizon-by-sector effects.

F Robustness of the Employment Effects to Noise in the Sector Elasticities

The sector elasticities in the main counterfactual are the raw 2-digit estimates from the stacked long-difference design of Appendix E. Part of their dispersion is sampling noise. I therefore re-solve the model with an empirical Bayes estimates, which shrink each 2-digit estimate toward the precision-weighted mean (0.847) in proportion to its sampling variance. Every other parameter is unchanged. After shrinkage, 61 of the 62 two-digit sectors are complements, compared with 36 with the raw estimates.

Table 14 shows that the aggregate employment effects barely change. For every 100 jobs created in de-regulated lines, the economy gains 56.9 jobs with the raw elasticities and 58.6 with the shrunk elasticities. The complementarity channel contributes 20.1 and 21.9 jobs, that is, 35 and 37 percent of the net effect. The scale and markup components, and the output gain, are essentially unchanged, since the elasticities enter only through the labor share of variable cost. Table 15 reports the full decomposition. What changes is which sectors the effects are attributed to: with the shrunk elasticities nearly all employment effects are in complement sectors, and never-reserved substitute sectors lose 0.8 jobs, compared with 16.2 with the raw elasticities.

The two common-elasticity rows separate the level of η from its dispersion. Imposing on every sector the baseline's employment-weighted mean elasticity, 0.86, gives 54.7 net jobs and a complementarity contribution of 17.4, so dispersion around the mean adds 2.2 jobs (4 percent) to the net effect. Imposing the pooled estimate of 0.76 instead raises the net effect to 64.9 and the complementarity contribution to 29.3: a lower average elasticity accounts for most of the difference between the baseline and the pooled benchmark. The aggregate complementarity effect thus depends mainly on the average elasticity of the sectors that buy the inputs made cheaper by the reform.

Targeted reforms

I also re-solve the targeted reforms of Table 9 and Figure 5 with the shrunk elasticities, keeping the packages fixed: they were chosen with the raw elasticities. Table 16 reports the two comparable reforms. With the raw elasticities, the reform chosen to reach substitutes creates 28 percent fewer

Table 14: Employment effect of de-reservation under alternative labour–materials elasticities

	2-digit sectors	Mean η	Net, economy-wide, per 100 jobs created in de-regulated lines				Never-reserved	Output gain
	$\eta < 1$		Scale	Markup	Compl.	Total	total	(%)
Baseline (raw estimates)	36/62	0.86	+26.7	+10.1	+20.1	+56.9	−43.1	+0.71
Shrunk (empirical Bayes)	61/62	0.83	+26.6	+10.1	+21.9	+58.6	−41.4	+0.71
Common $\eta = 0.76$	–	0.76	+25.7	+10.0	+29.3	+64.9	−35.1	+0.71
Common $\eta = 0.86$ (baseline mean)	–	0.86	+27.2	+10.1	+17.4	+54.7	−45.3	+0.71

Notes: Each row re-solves the main counterfactual (removing $\overline{K}$, entry and expansion) with a different vector of sector elasticities η and all other parameters unchanged. Employment columns follow Table 6: the economy-wide change in employment and its components per 100 jobs created in de-regulated lines, each specification normalised by its own de-regulated effect; Never-reserved total is the never-reserved sectors' contribution. 2-digit sectors $\eta < 1$ counts the 62 two-digit sectors with η below one; Mean η is weighted by baseline cell employment. Output gain as in Table 8. Baseline: stacked long-difference 2SLS estimates with the shift-share instrument, one-digit estimates where a two-digit estimate is unavailable. Shrunk: the same estimates shrunk toward their precision-weighted mean (0.847) by empirical Bayes, with the mean for sectors without an estimate.

Table 15: Decomposition of the employment effect of de-reservation, per 100 jobs created in de-regulated lines: long-difference sector elasticities shrunk by empirical Bayes

	Scale	Markup	Complementarity	Total
De-regulated lines				
$\eta < 1$	+86.0	+10.1	+3.2	+99.3
$\eta > 1$	+0.7	0.0	0.0	+0.7
All	+86.7	+10.1	+3.2	+100.0
Never-reserved sectors				
$\eta < 1$	−59.4	0.0	+18.7	−40.7
$\eta > 1$	−0.7	0.0	0.0	−0.8
All	−60.1	0.0	+18.7	−41.4
Net, economy-wide	+26.6	+10.1	+21.9	+58.6

Notes: Each entry is the change in employment attributed to a channel and group, per 100 jobs created in de-regulated lines (the de-regulated rows' total). Cells are three-digit sector × reservation status; de-regulated lines are the reserved cells of the 113 treated sectors, and never-reserved sectors include all other cells. The decomposition uses economy-wide logarithmic-mean (Sato–Vartia) weights, so rows are additive and the net row is the sum of the two group totals. Scale is the change in log revenue; Markup is minus the change in the log markup, measured as the revenue-weighted harmonic mean of firm markups in the cell, which equals the change in the log share of revenue spent on labour and materials; Complementarity is the change in the log labour share of variable cost. With the wage as numeraire the three sum to the change in log employment. η is the sector's labour–materials elasticity of substitution. The de-regulated lines' employment effect is 0.354 percent of baseline economy-wide employment (about 25,274 jobs at survey-expanded baseline employment); the economy-wide effect is 0.207 percent.

net jobs than the one chosen to reach complements. With the shrunk elasticities the gap in the effective elasticity of the buyers reached falls from 0.12 to 0.05, and the job gap to 11 percent; with a common elasticity it is 2 percent.

Table 17 and Figure 8 show the same for the 49 random pairs of Figure 5. The slope of the job gap on the gap in the effective elasticity of the buyers reached is similar with the raw and the shrunk elasticities (-4.2 and -3.3), while with a common elasticity job gaps are close to zero. Shrinkage compresses the gaps, but preserves their ranking: the correlation of the pairs' job gaps with those under the raw elasticities is 0.93, and 86 percent of the pairs keep their sign. Heterogeneity in η therefore matters for the ranking of reforms once sampling noise is removed, although the size of the differences between specific reforms is smaller than the raw estimates imply.

Table 16: Two comparable reforms under alternative labour–materials elasticities

	Raw C / S	Shrunk C / S	Common $\eta = 0.76$ C / S
Effective elasticity of the buyers reached	0.86 / 0.97	0.82 / 0.87	0.76 / 0.76
Net jobs, per 100 workers in de-regulated sectors	1.1 / 0.8	1.2 / 1.1	1.4 / 1.4
Net jobs, per 100 created in de-regulated sectors	43.3 / 30.1	48.6 / 42.0	58.2 / 55.0
Labor–materials channel, per 100 created	22.3 / 3.2	27.9 / 16.1	38.0 / 30.1
Gap in net jobs, S vs. C	-28%	-11%	-2%

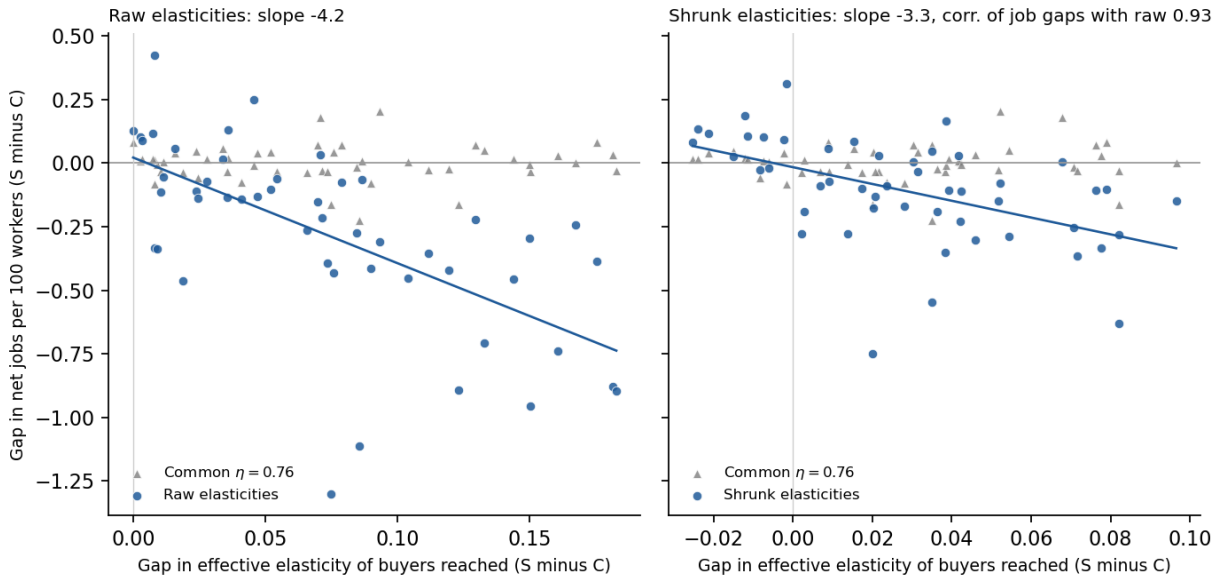
Notes: The two reforms of Table 9 (C: the package chosen to reach complements; S: the package chosen to reach substitutes, both under the raw elasticities), each re-solved jointly with every vector of sector elasticities and all other parameters unchanged. Raw: stacked long-difference 2SLS estimates (the baseline). Shrunk: the same estimates shrunk by empirical Bayes. The effective elasticity of the buyers reached uses each column's elasticities and weights each cell's elasticity by its employment, its materials share of labor and materials costs, and the fall in its materials price. Jobs are the economy-wide employment gain per 100 workers employed in the de-regulated sectors before the reform; the gap is the percentage difference between S and C.

Table 17: Random policy pairs under alternative labour–materials elasticities

	S.d. across pairs of the gap in		Slope	Job gap vs. raw	
	effective η	net jobs		Correlation	Same sign
Raw	0.054	0.35	-4.2	1.00	100%
Shrunk	0.031	0.21	-3.3	0.93	86%
Common $\eta = 0.76$	–	0.07	–	–	–

Notes: The 49 fixed-rule pairs of Figure 5, each package re-solved jointly with every vector of sector elasticities. Gaps are S minus C, with C the package whose buyers have the lower effective elasticity under the raw elasticities, kept fixed across rows. Net jobs per 100 workers employed in the de-regulated sectors before the reform. Slope: OLS slope of the job gap on the gap in effective elasticity. Job gap vs. raw: correlation across pairs with the job gaps under the raw elasticities, and the share of pairs with the same sign.

Figure 8: Elasticity Gap of Reached Sectors vs Job Creation with Shrunk Elasticities



Notes: Each point is one of the 49 random pairs of Figure 5, with both packages re-solved under the elasticities named in the panel (circles) and under a common $\eta = 0.76$ (triangles). S is the package whose buyers have the higher effective elasticity under the raw elasticities, kept fixed across panels, so gaps on the horizontal axis can be negative. The line is the OLS fit.

G Input Dereservation: Alternative Treatment Dates

The input event study in Section 1.3 dates the treatment of a firm by the year in which its main input is dereserved. Because firms use several inputs and change their input baskets over time, this date can be set in more than one way. This appendix re-estimates the event study under three definitions and shows that the main result, that employment rises after input dereservation in complement sectors but not in substitute sectors, does not depend on the choice.

G.1 Treatment dates

Let $\mathcal{M}_{it}$ be the set of inputs that firm i reports in year t, T_m the year in which input m is removed from the reservation list, and t_{0i} the first year in which firm i is observed. I consider three treatment

dates:

$$E_i^A = \min_t \min_{m \in \mathcal{M}_{it}} T_m,$$

$$E_i^U = \min \{T_m : \ m \in \mathcal{M}_{it} \text{ for some } t \leq T_m\},$$

$$E_i^M = T_{m_i^*}, \qquad m_i^* = \text{the input with the largest expenditure share in } \mathcal{M}_{it_{0i}}.$$

Any input, E_i^A, is the broadest definition: the firm is treated from the first dereservation of any input it ever uses. This includes inputs the firm adopts only after they are dereserved, which dates the treatment by a purchase decision that may itself respond to the policy. *Input in use*, E_i^U, counts an input only if the firm already uses it in or before the year it is dereserved. *Main input*, E_i^M, is the definition used in the main text: it uses only the dereservation of the firm's largest input in its first year, which is fixed before the reform reaches the firm and carries the largest share of its input spending. All three dates are computed on the full input-level panel, before the sample restrictions. Under E_i^U and E_i^M, firms that are treated under E_i^A but not under the alternative date are dropped rather than used as controls. The date differs from E_i^A for 2,178 firms under E_i^U and for 1,558 firms under E_i^M.

The sample, the outcome variables, and the exclusion of firms affected through the output side are those of the main text. Sectors are classified as complements if $\eta_s < 1$ and substitutes if $\eta_s > 1$, using the 2-digit elasticities of Appendix E (the 1-digit estimate where the 2-digit one is not available).

G.2 Estimators

For each date I estimate the event study with three estimators. The first is the two-way fixed effects specification of the main text, with event-time indicators interacted with the sector type and firm and year fixed effects. The second is the interaction-weighted estimator of Sun and Abraham (2021), which estimates a separate path for each dereservation cohort and sector type and aggregates them with cohort shares, so that already-treated firms never serve as controls. The third is a stacked difference-in-differences (Cengiz et al., 2019): for each cohort g I build a data set with years $g-3$ to $g+4$, the firms first treated in g, and clean controls (firms never treated and firms first treated after

$g + 4$), and estimate the event study on the stacked data with firm-by-stack and year-by-stack fixed effects. Event time runs from -3 to 4; in the first two estimators the end points bin all earlier and later years. The omitted year is $k = -1$, and standard errors are clustered by firm.

I also estimate a static version that replaces the event-time indicators with a single post-treatment indicator,

$$\log y_{it} = a \, \text{Post}_{it} + d \, \text{Post}_{it} \times D_s^S + \alpha_i + \delta_t + \varepsilon_{it},$$

where a is the effect in complement sectors and $a + d$ the effect in substitute sectors.

G.3 Results

Table 18 reports the average of the event-time coefficients for $k = 0, \ldots, 4$, and Figures 9–**??** show the full paths. Employment in complement sectors rises by 4 to 7 log points after input dereservation, and the gap with substitute sectors is positive and significant under every date and estimator. The gap is largest when the treatment is dated by the main input (0.10 to 0.12), consistent with a larger shock to input costs. In substitute sectors employment does not increase; under all three dates the point estimates are negative and mostly insignificant. For complements, the rejection of the joint pre-trend test under the first two dates comes from the binned end point $k \leq -3$ (between -0.02 and -0.04); the coefficient at $k = -2$ is close to zero under every date and estimator. Under the main-input date the TWFE leads are jointly insignificant for both sector types ($p = 0.16$ for complements and 0.20 for substitutes).

Intermediate input quantities rise in both sector types, and the gap between them is not significant. In substitute sectors, input quantities are already rising before dereservation (the coefficient at $k = -2$ is negative relative to $k = -1$), and the leads are jointly significant under every date, so I do not interpret the input-quantity results for substitutes as causal.

Table 19 reports the static estimates. Employment rises by 3 to 6 log points in complement sectors, and the difference for substitute sectors (d) is negative and significant in every column, so the total effect for substitute sectors is close to zero. A specification with a continuous interaction $\text{Post}_{it} \times (\eta_s - 1)$ yields small and insignificant slopes: the employment response changes at $\eta_s \approx 1$ rather than linearly in η_s.

Table 18: Input Event Study by Treatment Date: Post-Treatment Averages

		Complements	Substitutes	C − S
Log workers				
Any input	TWFE	0.056***	-0.023	0.079***
		(0.011)	(0.017)	(0.019)
	Sun–Abraham	0.060***	-0.026	0.086***
		(0.011)	(0.017)	(0.020)
	Stacked	0.054***	-0.014	0.068***
		(0.011)	(0.017)	(0.019)
Input in use	TWFE	0.047***	-0.032*	0.079***
		(0.012)	(0.019)	(0.022)
	Sun–Abraham	0.051***	-0.034*	0.084***
		(0.012)	(0.019)	(0.022)
	Stacked	0.044***	-0.026	0.069***
		(0.012)	(0.019)	(0.022)
Main input	TWFE	0.071***	-0.036	0.107***
		(0.016)	(0.025)	(0.029)
	Sun–Abraham	0.073***	-0.049*	0.122***
		(0.016)	(0.026)	(0.030)
	Stacked	0.066***	-0.034	0.099***
		(0.016)	(0.024)	(0.029)
Log intermediate input quantity				
Any input	TWFE	0.108***	0.150***	-0.042
		(0.035)	(0.057)	(0.064)
	Sun–Abraham	0.107***	0.187***	-0.080
		(0.035)	(0.058)	(0.065)
	Stacked	0.120***	0.159***	-0.039
		(0.035)	(0.055)	(0.063)
Input in use	TWFE	0.111***	0.132**	-0.021
		(0.040)	(0.064)	(0.073)
	Sun–Abraham	0.100***	0.163**	-0.063
		(0.038)	(0.065)	(0.073)
	Stacked	0.124***	0.136**	-0.011
		(0.040)	(0.062)	(0.072)
Main input	TWFE	0.095*	0.157*	-0.062
		(0.052)	(0.094)	(0.106)
	Sun–Abraham	0.085*	0.190*	-0.105
		(0.050)	(0.097)	(0.108)
	Stacked	0.117**	0.101	0.016
		(0.052)	(0.091)	(0.103)

Notes: Average of the event-time coefficients for $k = 0, \ldots, 4$, relative to $k = -1$, for firms in complement ($\eta_s < 1$) and substitute ($\eta_s > 1$) sectors, and their difference. Treatment dates are defined in the text. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

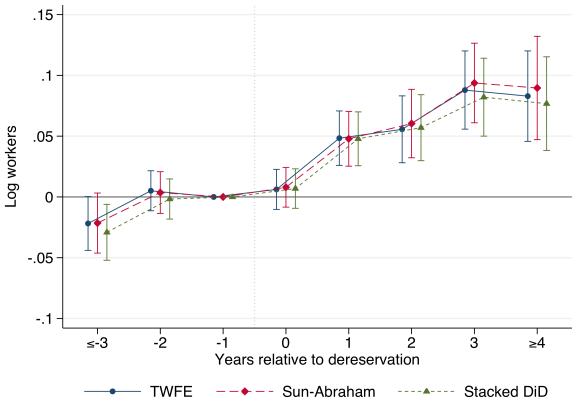
Table 19: Static Difference-in-Differences by Treatment Date

	Any input		Input in use		Main input	
	TWFE	Stacked	TWFE	Stacked	TWFE	Stacked
Log workers						
Post	0.036***	0.044***	0.028***	0.032***	0.053***	0.056***
	(0.009)	(0.009)	(0.009)	(0.010)	(0.014)	(0.014)
Post $\times$ 1($\eta > 1$)	-0.051***	-0.041**	-0.052***	-0.044***	-0.092***	-0.074***
	(0.017)	(0.016)	(0.017)	(0.017)	(0.024)	(0.023)
Post, $\eta > 1$ (sum)	-0.014	0.003	-0.023	-0.011	-0.039*	-0.019
	(0.014)	(0.014)	(0.015)	(0.015)	(0.021)	(0.020)
Post $\times$ ($\eta - 1$)	-0.026	-0.007	-0.007	0.006	-0.042	-0.006
	(0.031)	(0.030)	(0.030)	(0.029)	(0.043)	(0.040)
Treated firms	9,334	7,549	7,584	5,963	4,651	3,569
Log intermediate input quantity						
Post	0.134***	0.135***	0.148***	0.133***	0.119***	0.109**
	(0.029)	(0.030)	(0.031)	(0.032)	(0.044)	(0.044)
Post $\times$ 1($\eta > 1$)	-0.065	0.001	-0.117**	-0.054	-0.232***	-0.128*
	(0.054)	(0.053)	(0.058)	(0.056)	(0.080)	(0.078)
Post, $\eta > 1$ (sum)	0.069	0.135***	0.032	0.079	-0.112*	-0.019
	(0.047)	(0.046)	(0.050)	(0.049)	(0.068)	(0.067)
Post $\times$ ($\eta - 1$)	0.043	0.095	0.005	0.035	0.030	0.125
	(0.095)	(0.092)	(0.102)	(0.099)	(0.157)	(0.150)
Treated firms	9,334	7,549	7,584	5,963	4,651	3,569

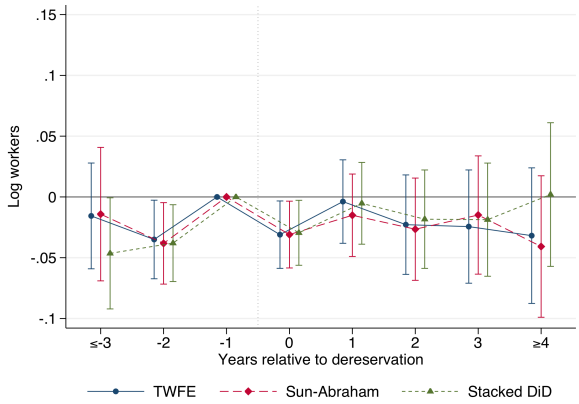
Notes: The first three rows of each panel come from the binary specification in the text: Post (effect in complement sectors), Post $\times$ 1($\eta_s > 1$) (difference for substitute sectors), and their sum. The fourth row replaces the sector-type interaction with $\text{Post}_{it} \times (\eta_s - 1)$. TWFE uses the full panel with firm and year fixed effects; Stacked uses the stacked data with firm-by-stack and year-by-stack fixed effects. Treated firms counts the treated firms in the estimation sample. Standard errors clustered by firm in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 9: Employment after Input Dereservation: Any Input

(a) Complements ($\eta_s < 1$)



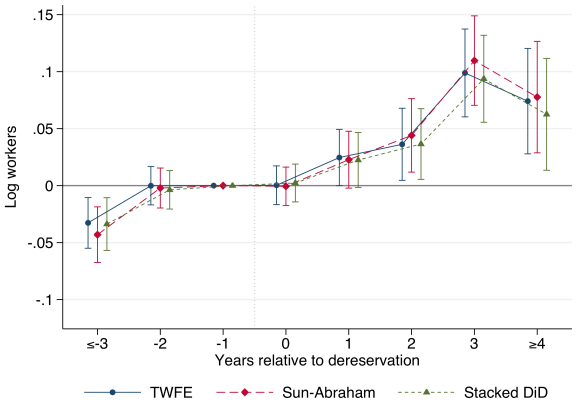
(b) Substitutes ($\eta_s > 1$)



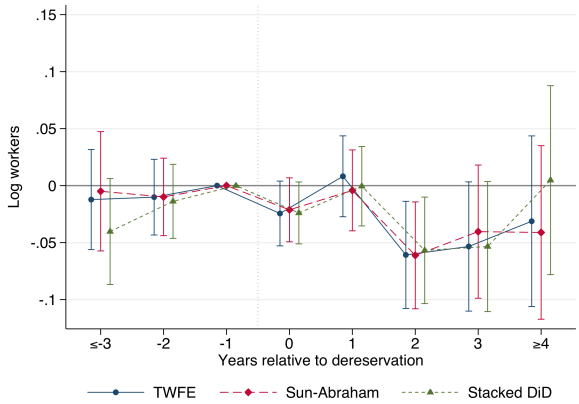
Notes: Event-time coefficients relative to $k = -1$ with 95% confidence intervals; standard errors clustered by firm. TWFE and Sun–Abraham bin all years before -3 and after 4 in the end points; the stacked window runs from $g - 3$ to $g + 4$. Post-treatment averages are in Table 18.

Figure 10: Employment after Input Dereservation: Input in Use

(a) Complements ($\eta_s < 1$)



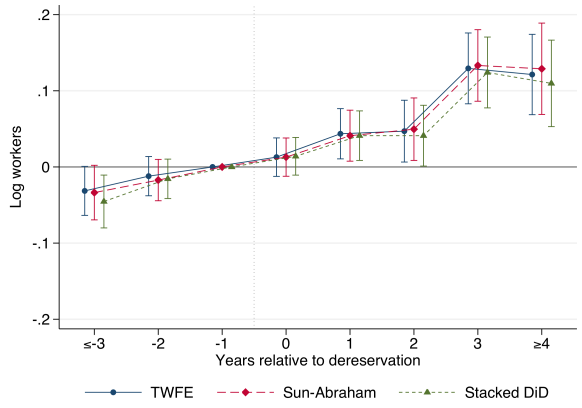
(b) Substitutes ($\eta_s > 1$)



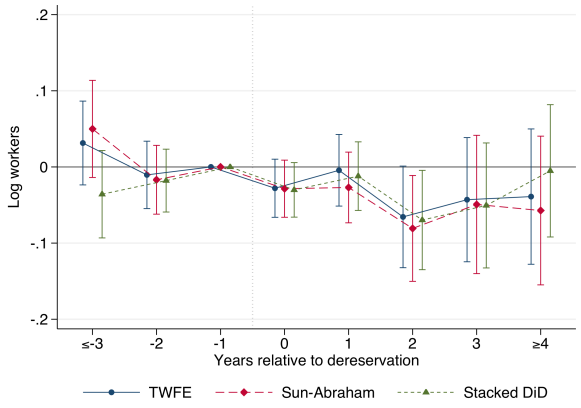
Notes: Event-time coefficients relative to $k = -1$ with 95% confidence intervals; standard errors clustered by firm. TWFE and Sun–Abraham bin all years before -3 and after 4 in the end points; the stacked window runs from $g - 3$ to $g + 4$. Post-treatment averages are in Table 18.

Figure 11: Employment after Input Dereservation: Main Input

(a) Complements ($\eta_s < 1$)



(b) Substitutes ($\eta_s > 1$)



Notes: Event-time coefficients relative to $k = -1$ with 95% confidence intervals; standard errors clustered by firm. TWFE and Sun–Abraham bin all years before -3 and after 4 in the end points; the stacked window runs from $g - 3$ to $g + 4$. Post-treatment averages are in Table 18.